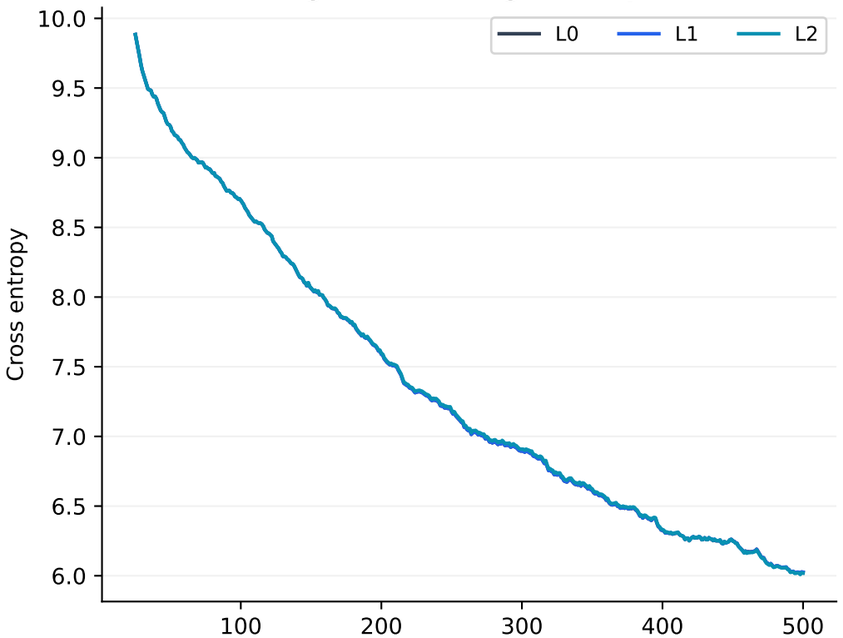
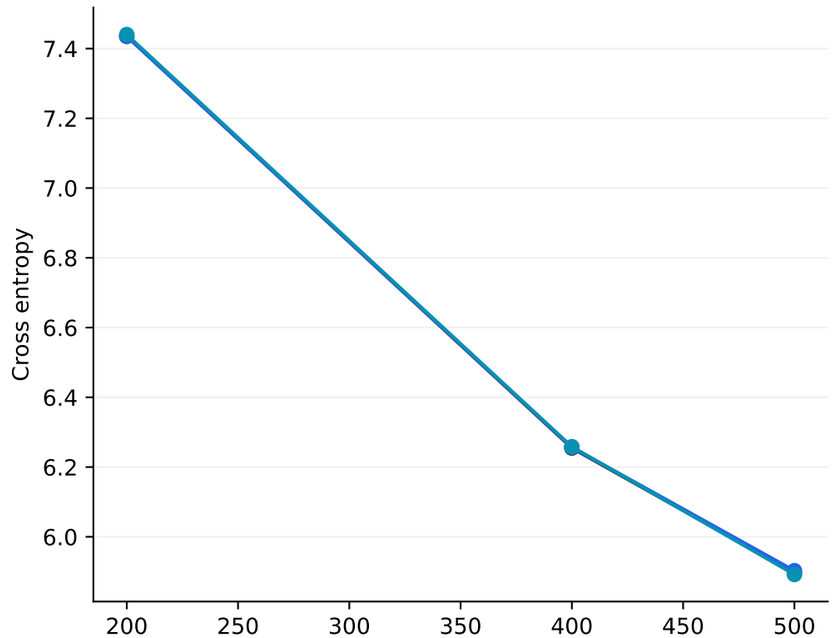
lion | 124M Llama | packed states | 500 steps | seed 0

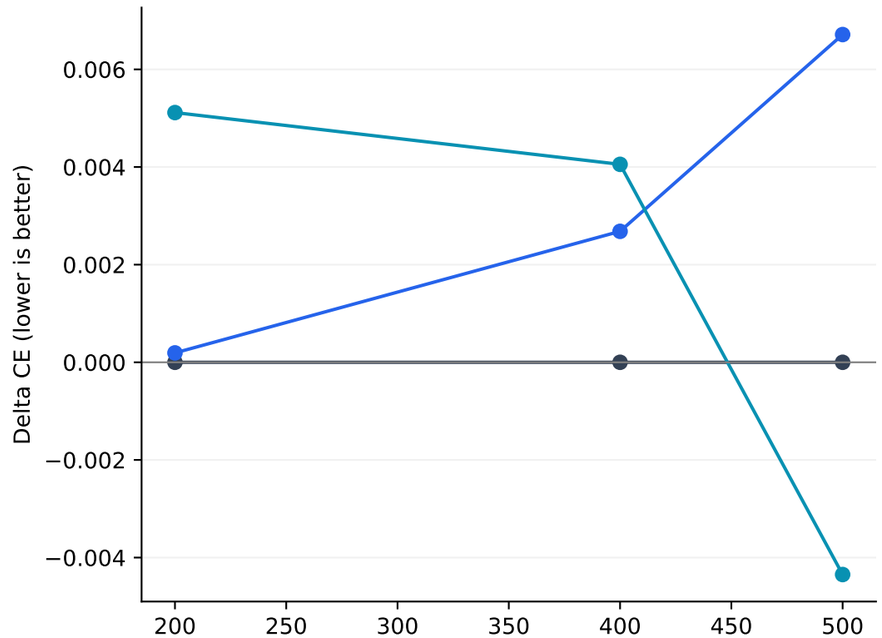
Training loss: trailing 25-step mean



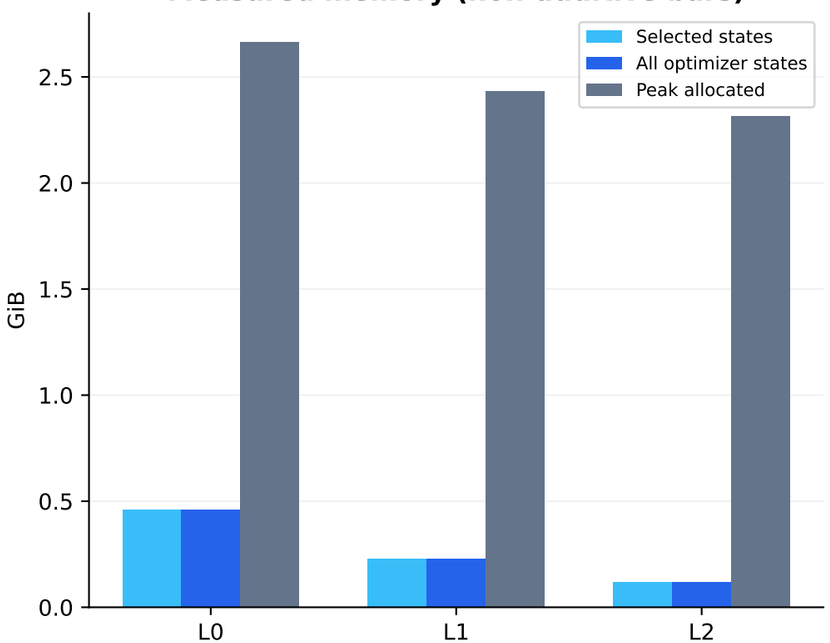
Validation loss: all 3 observations



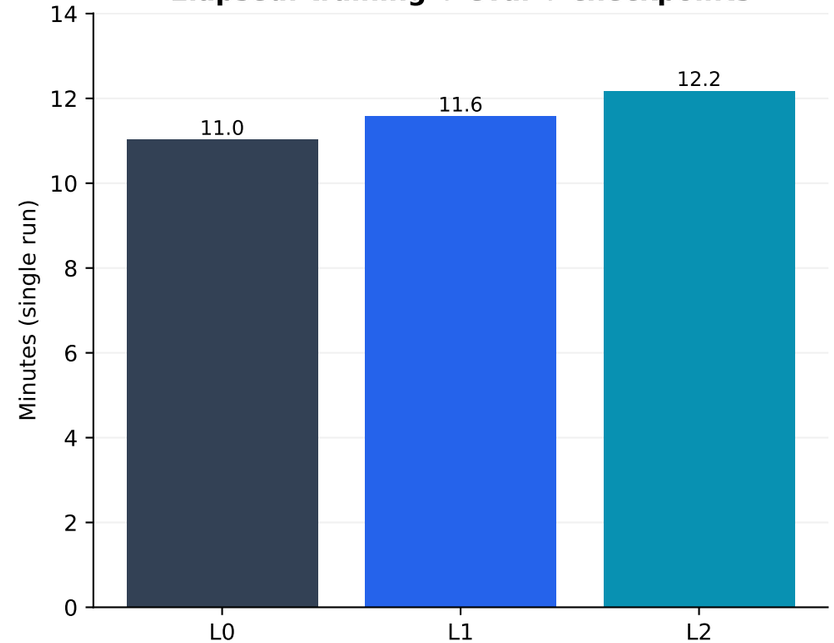
Validation difference vs FP32 baseline



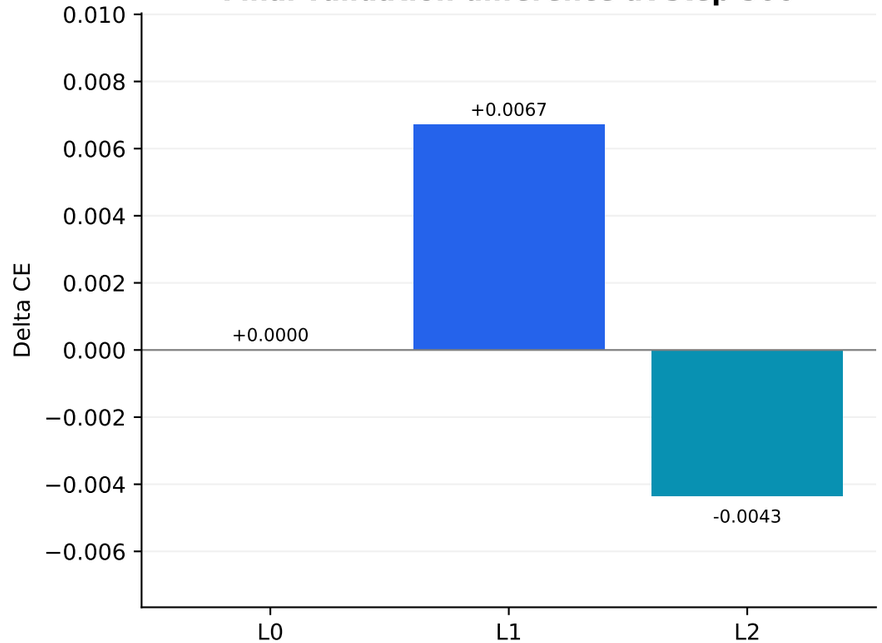
Measured memory (non-additive bars)



Elapsed: training + eval + checkpoints

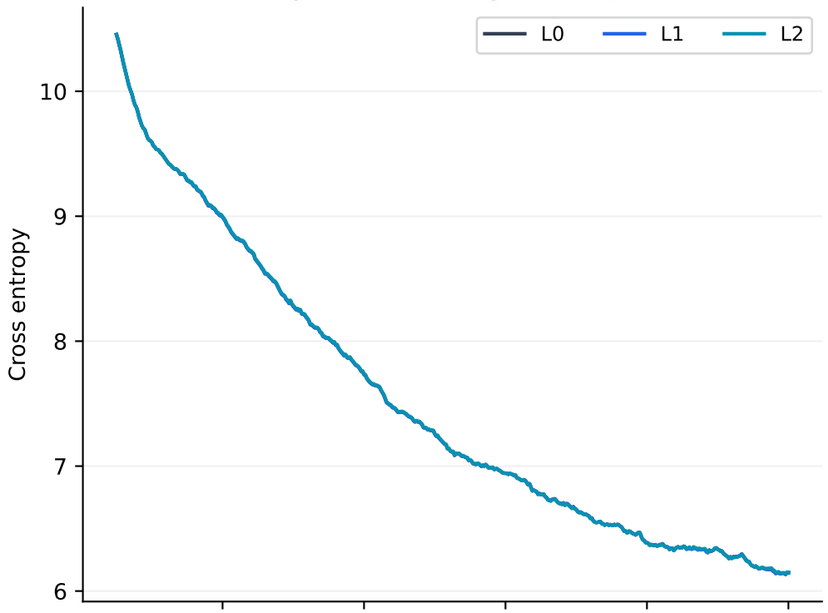


Final validation difference at step 500

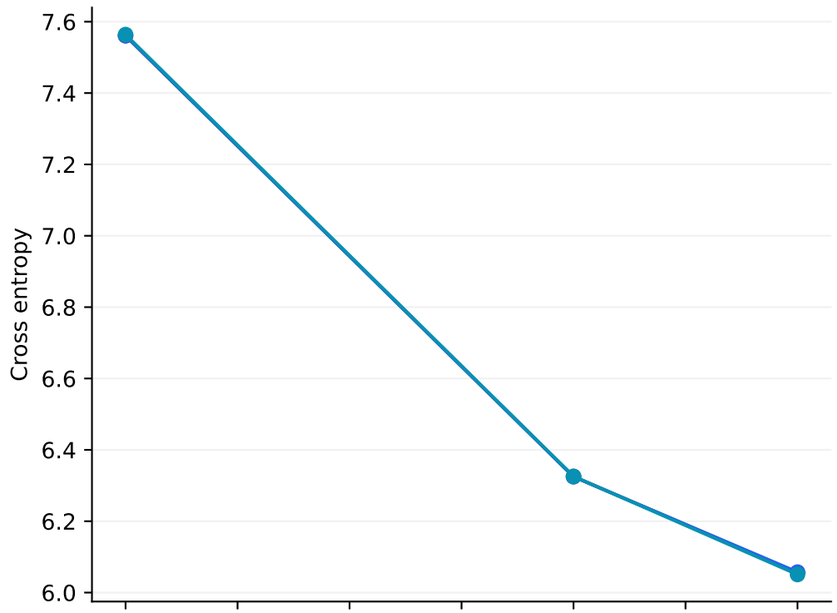


signum | 124M Llama | packed states | 500 steps | seed 0

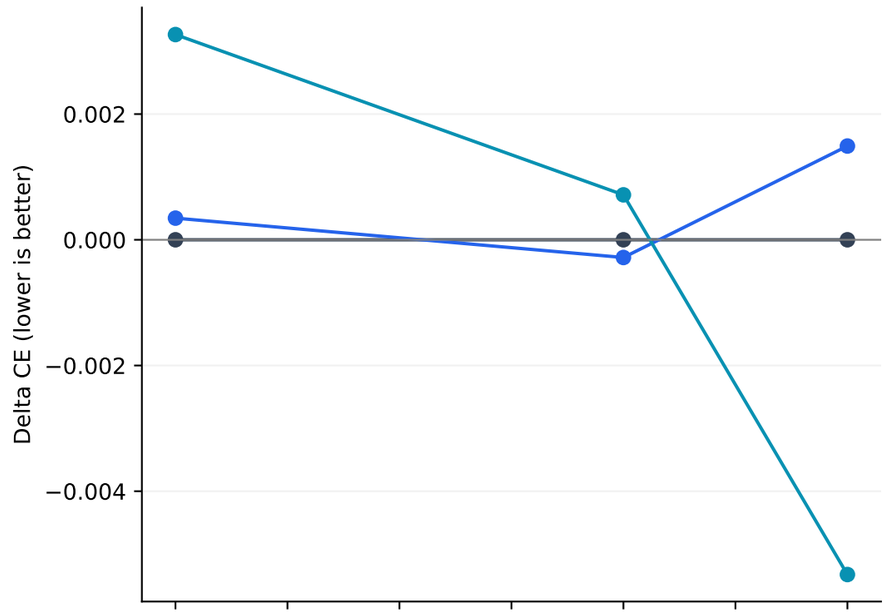
Training loss: trailing 25-step mean



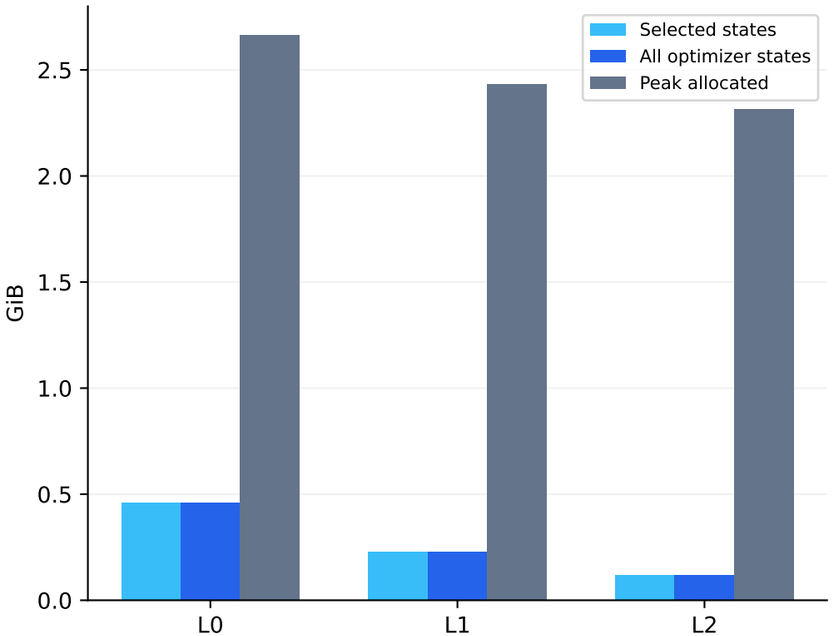
Validation loss: all 3 observations



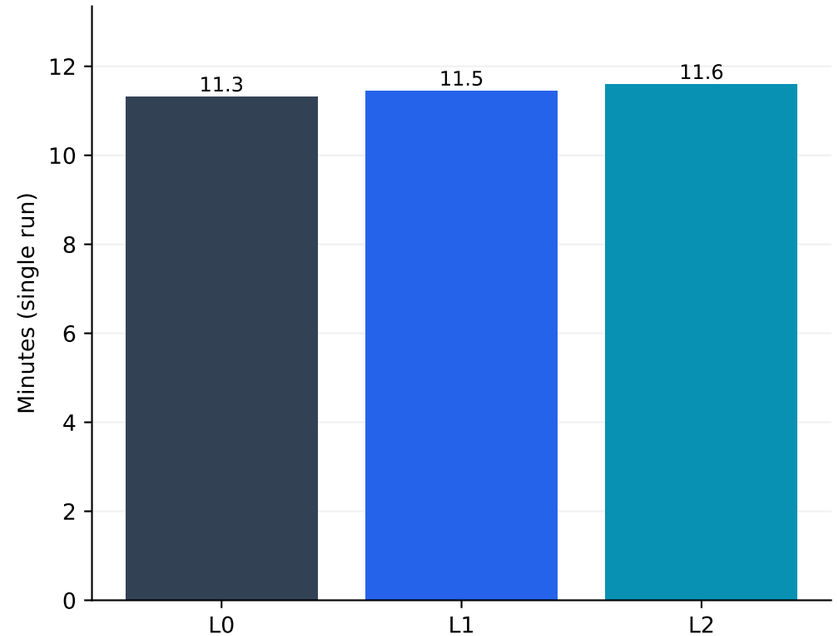
Validation difference vs FP32 baseline



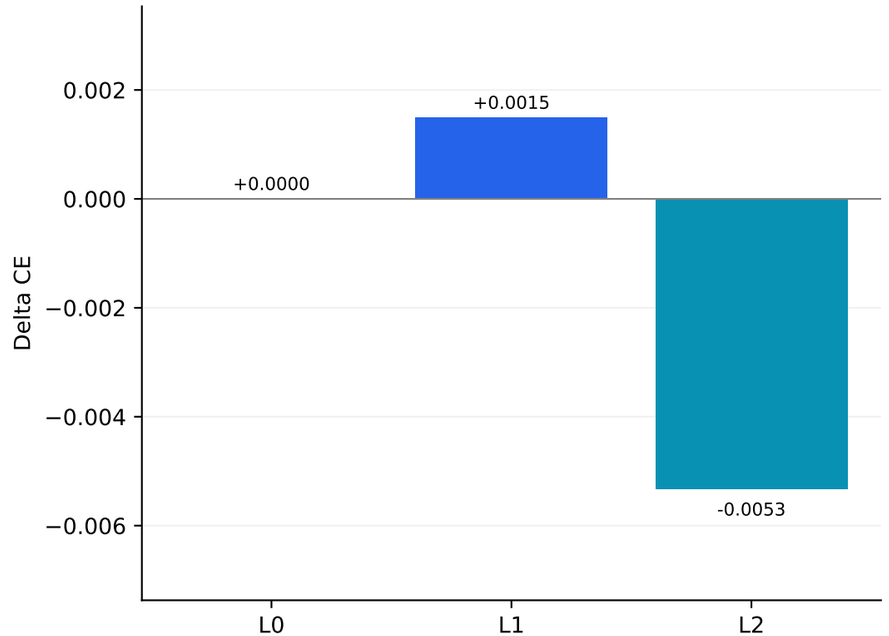
Measured memory (non-additive bars)



Elapsed: training + eval + checkpoints

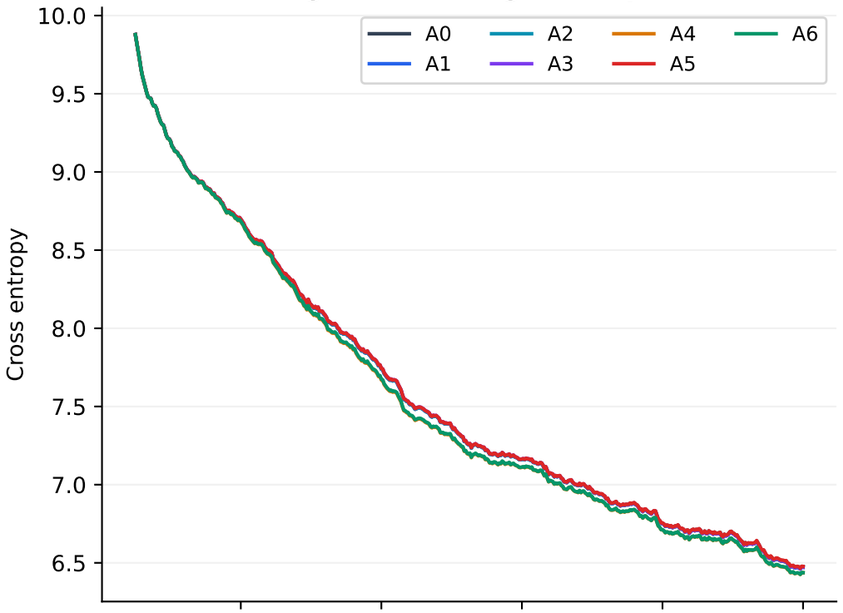


Final validation difference at step 500

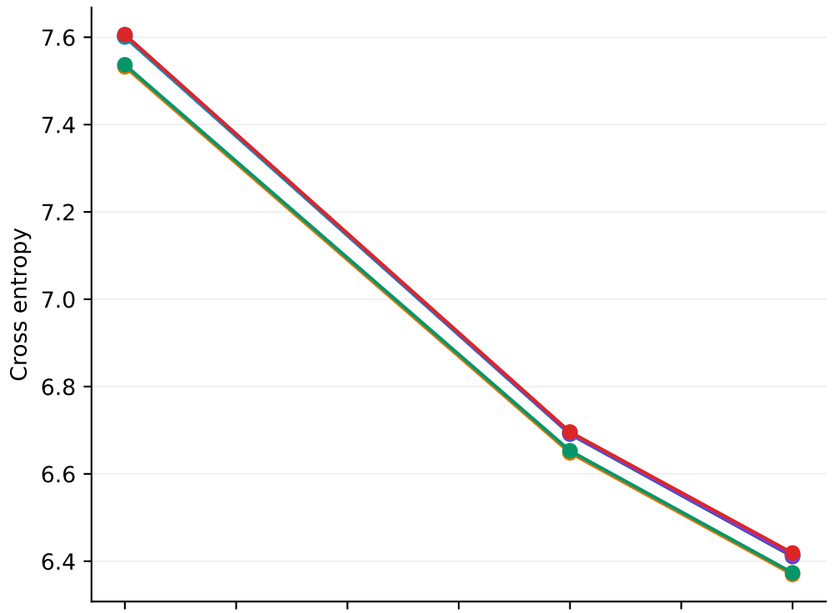


cadamw | 124M Llama | packed states | 500 steps | seed 0

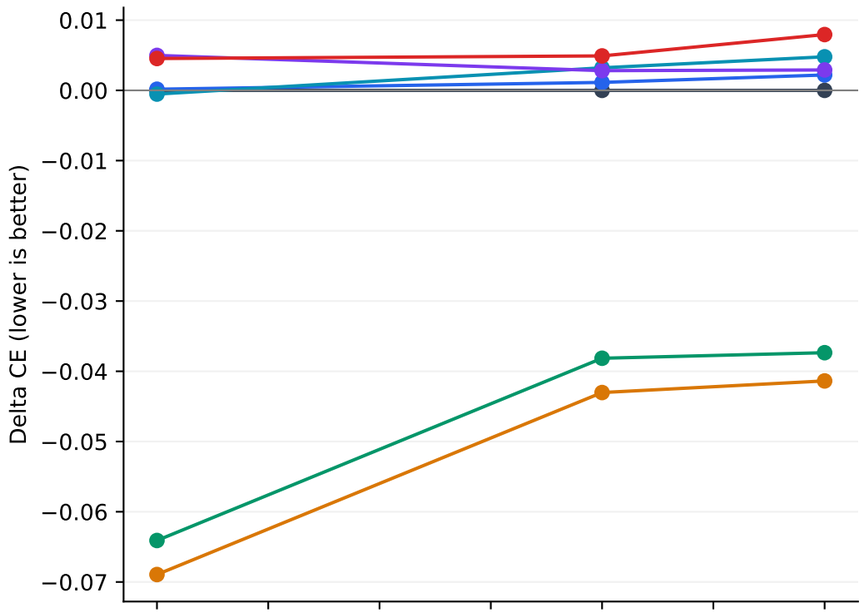
Training loss: trailing 25-step mean



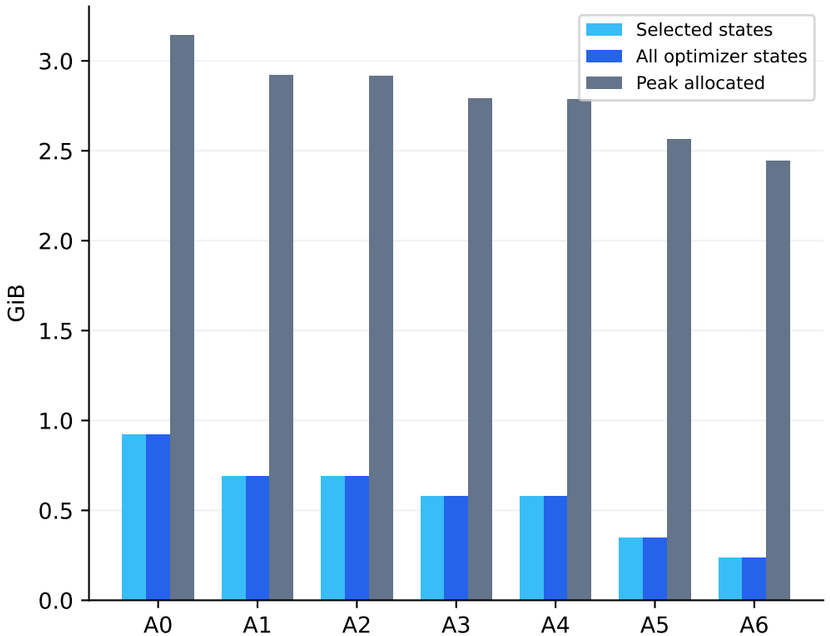
Validation loss: all 3 observations



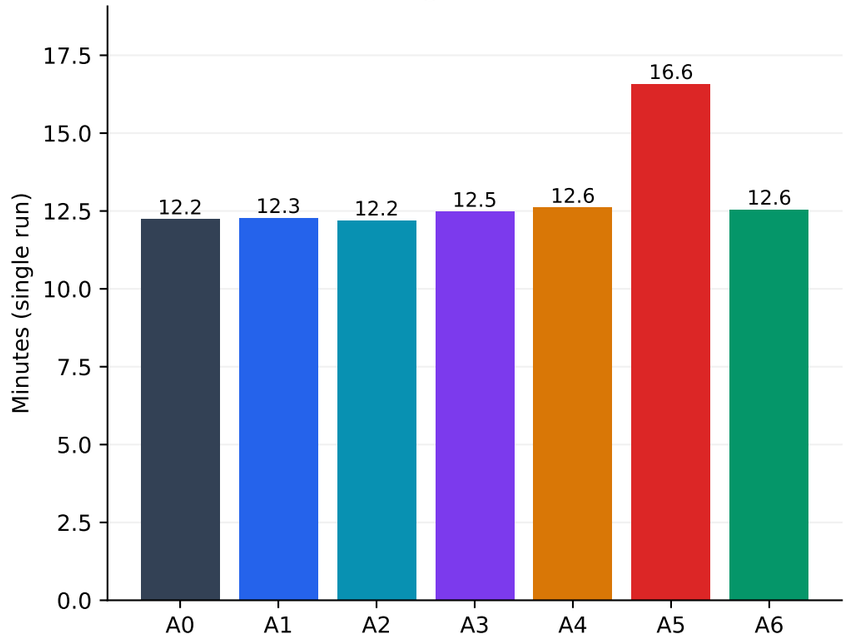
Validation difference vs FP32 baseline



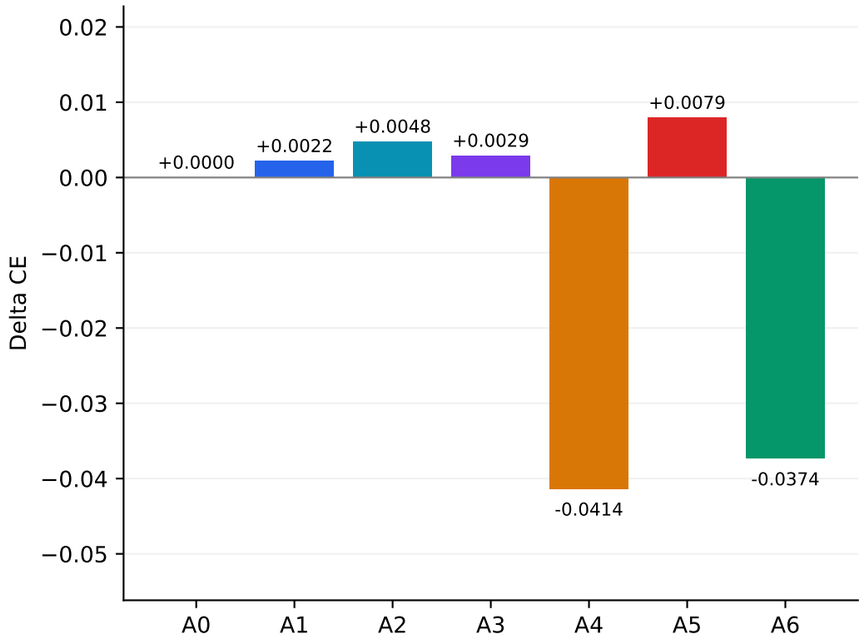
Measured memory (non-additive bars)



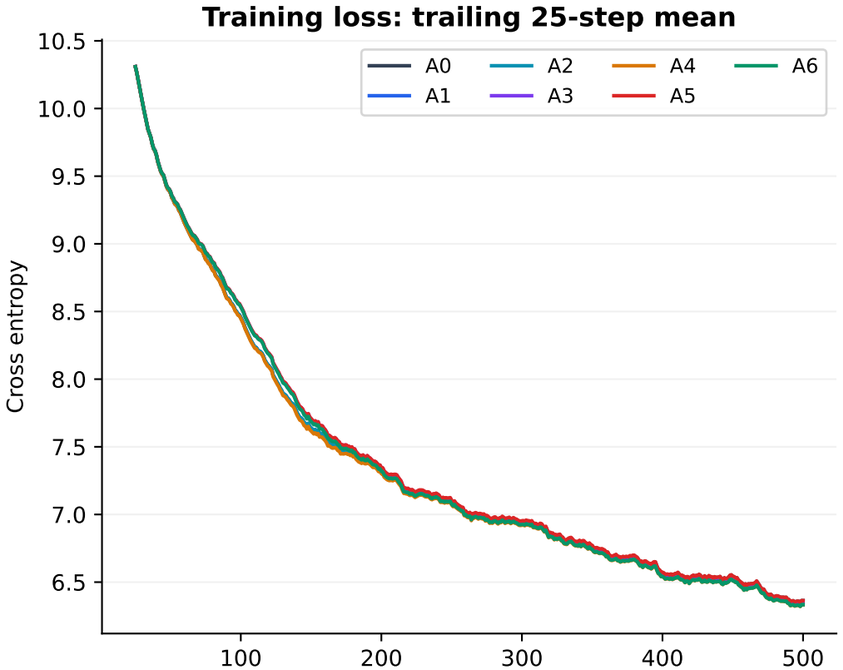
Elapsed: training + eval + checkpoints

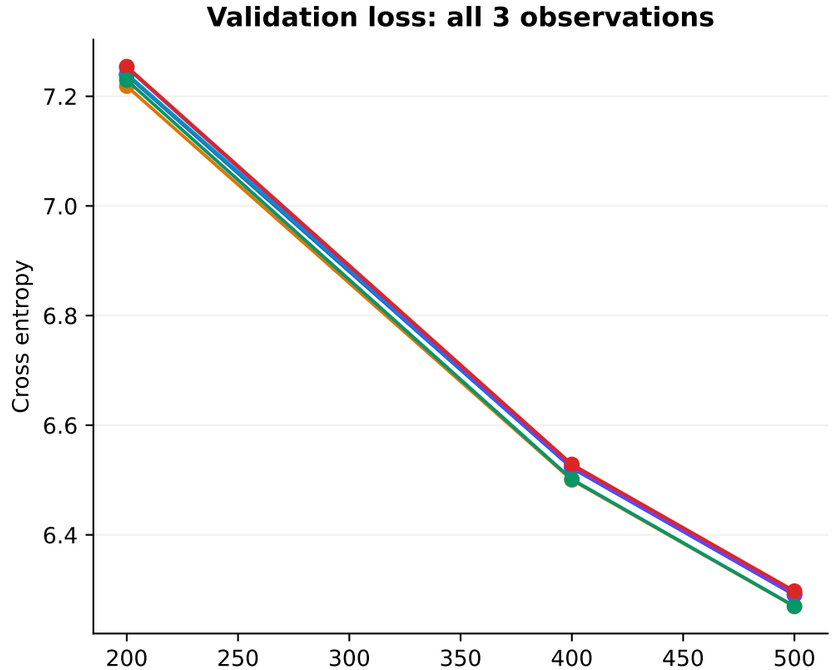


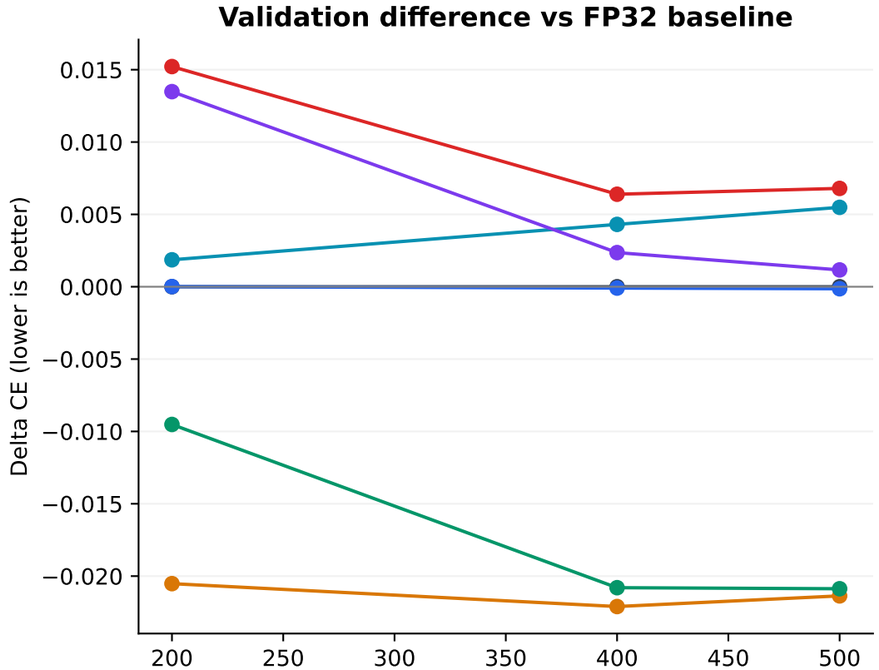
Final validation difference at step 500

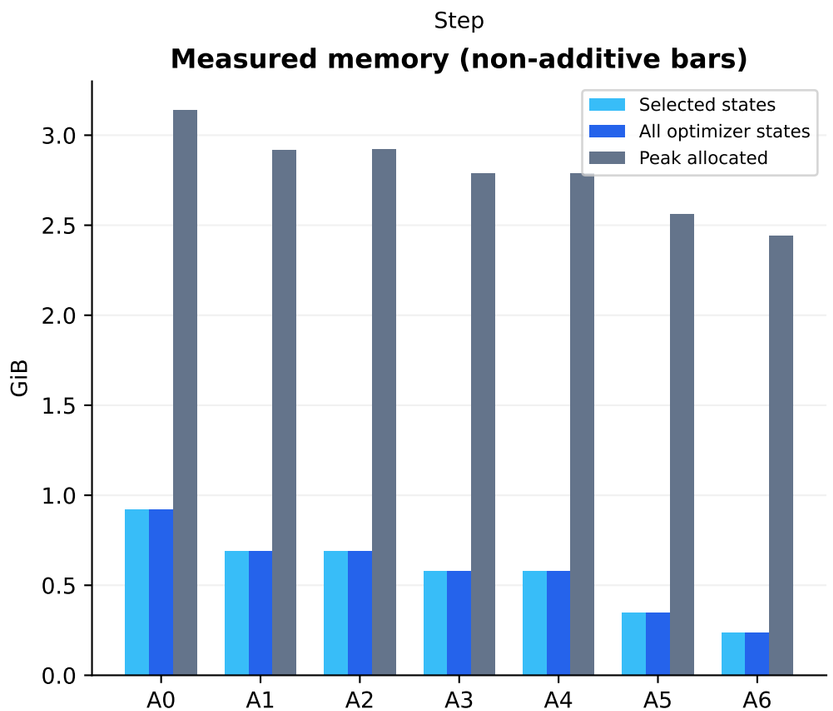


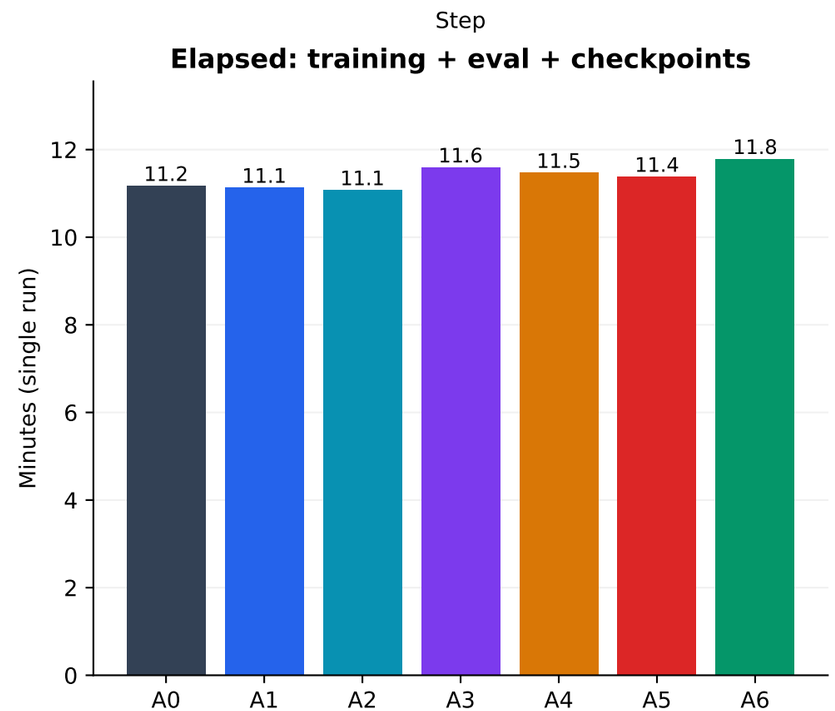
adopt | 124M Llama | packed states | 500 steps | seed 0

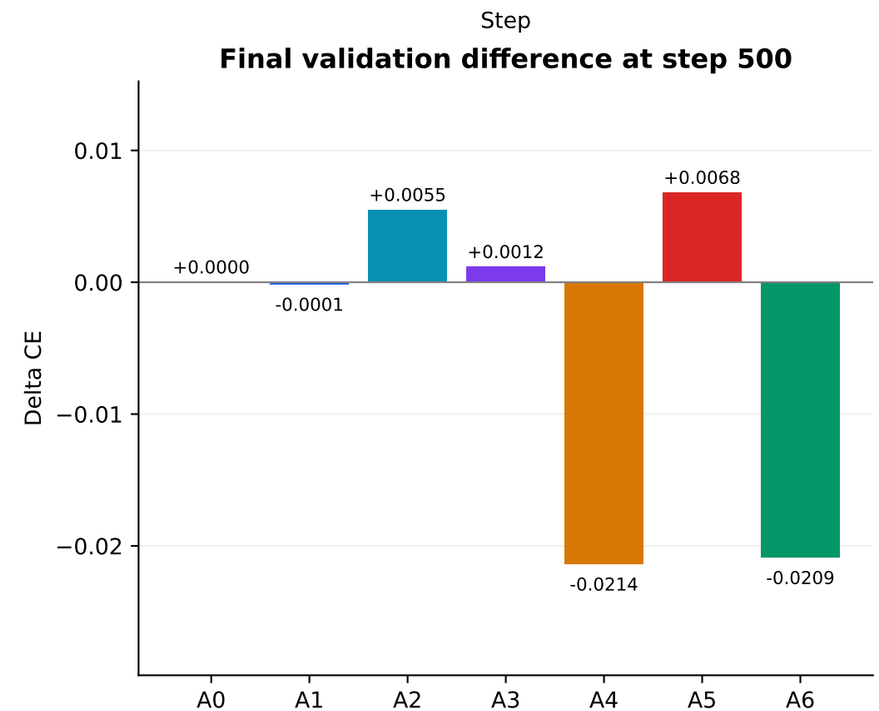






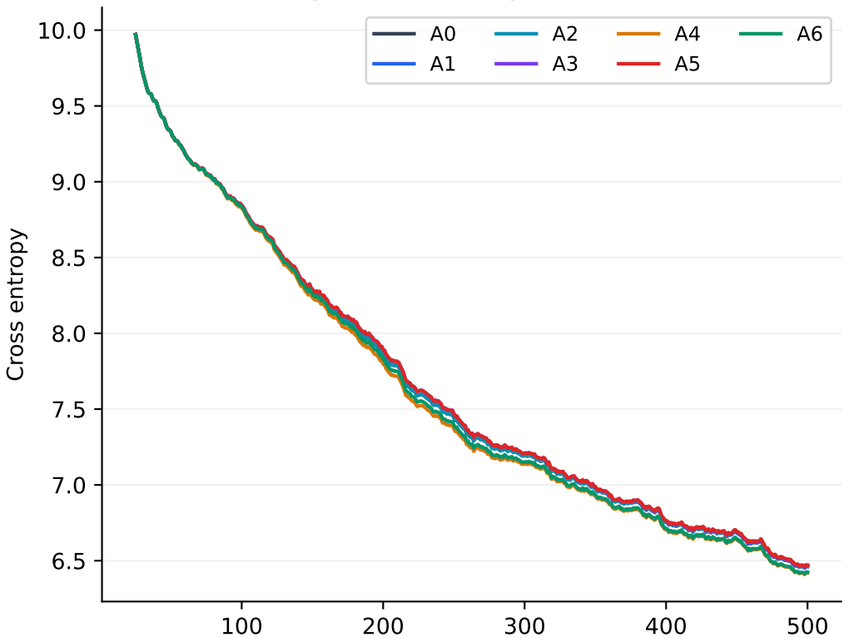




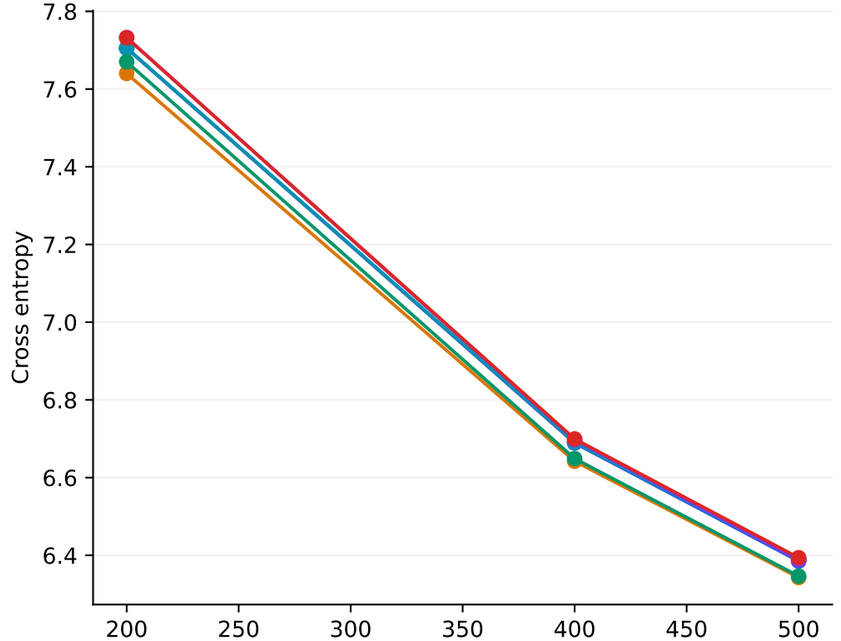


ademamix | 124M Llama | packed states | 500 steps | seed 0

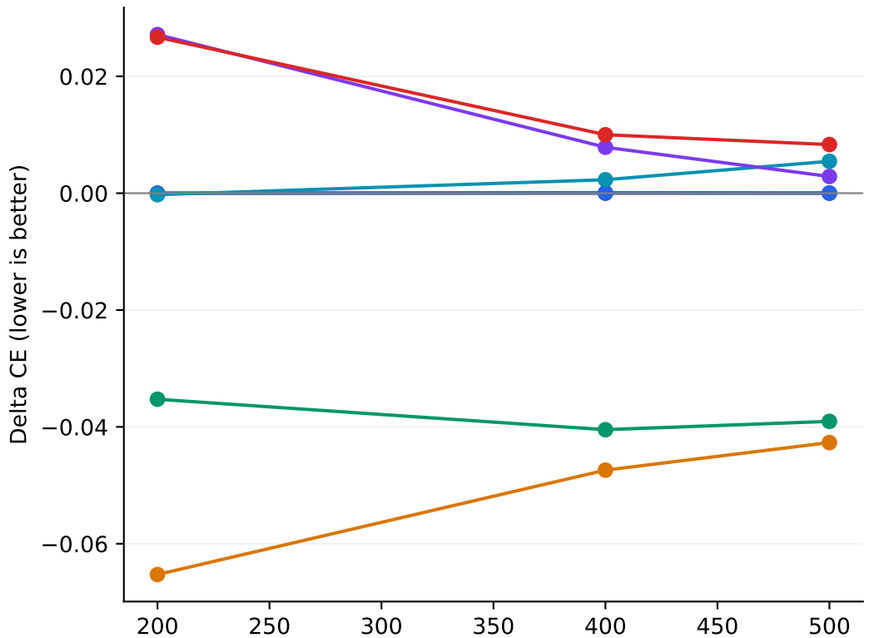
Training loss: trailing 25-step mean



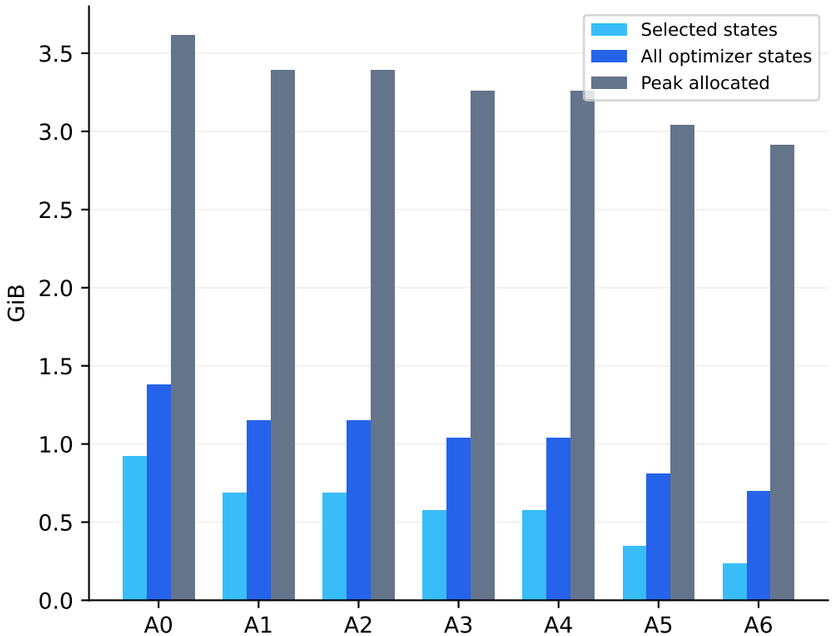
Validation loss: all 3 observations



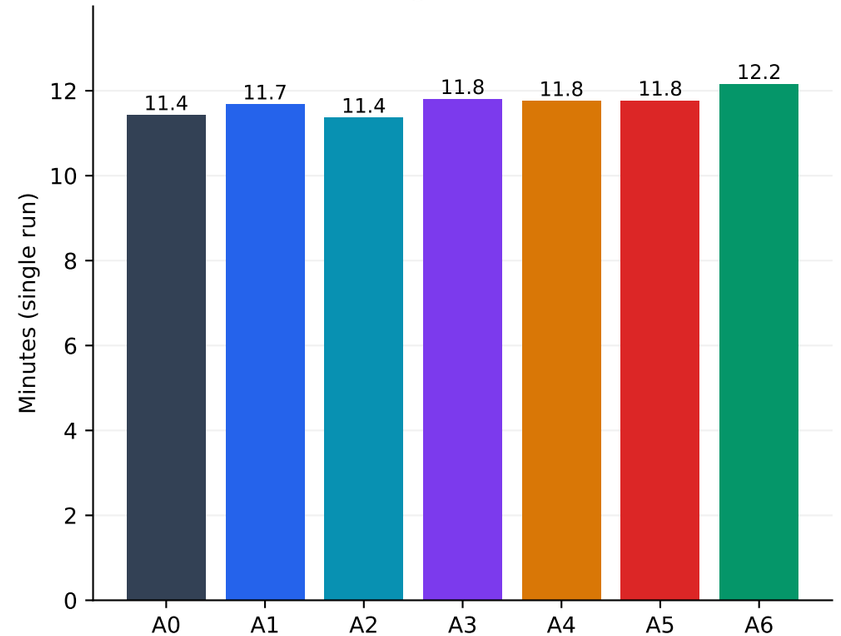
Validation difference vs FP32 baseline



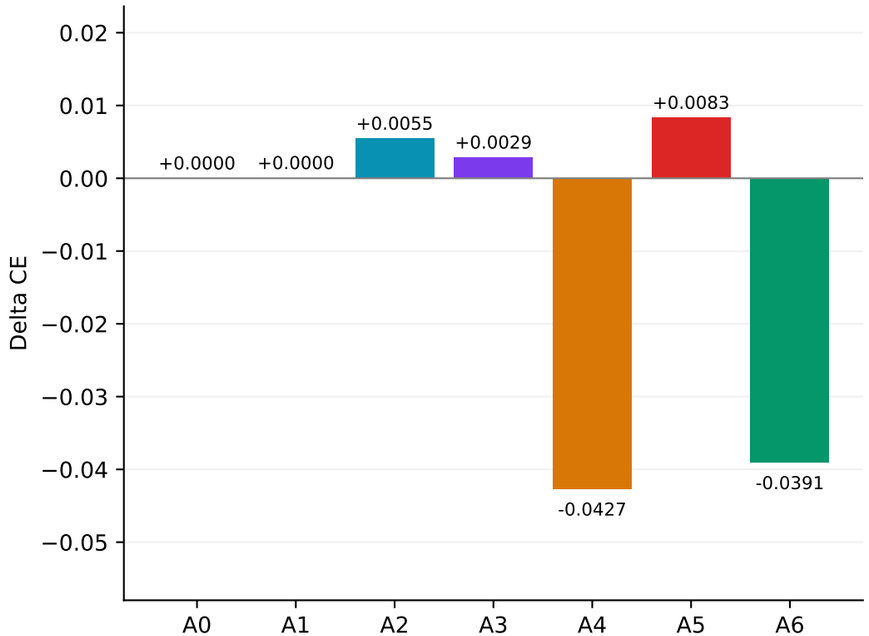
Measured memory (non-additive bars)



Elapsed: training + eval + checkpoints

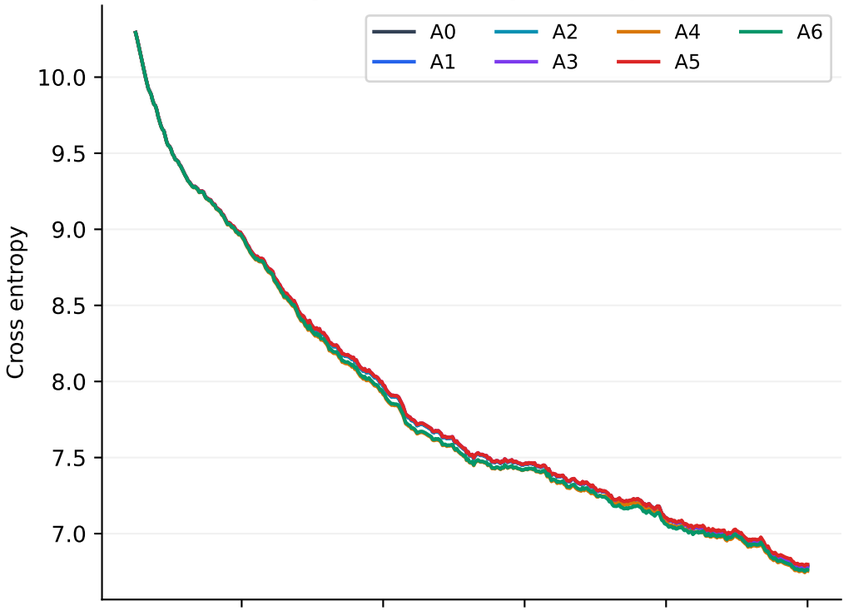


Final validation difference at step 500

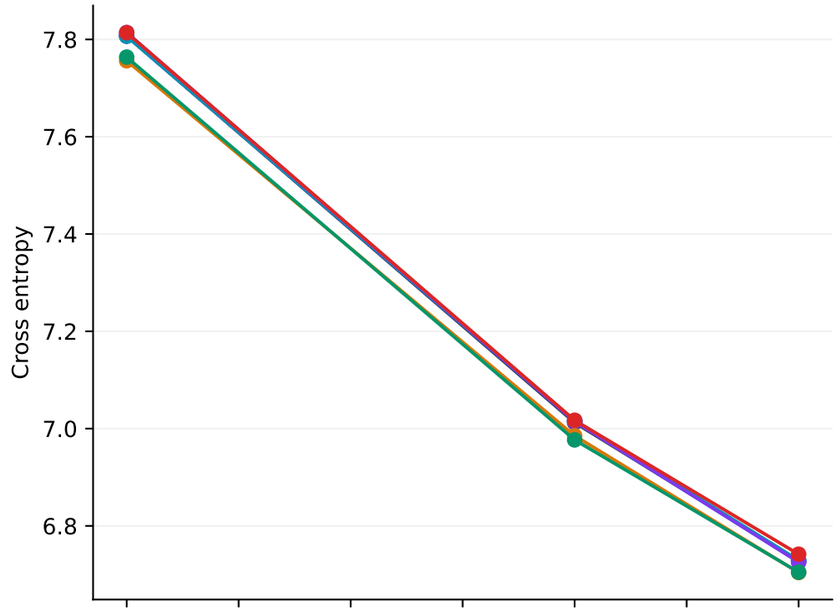


adamw-magma | 124M Llama | packed states | 500 steps | seed 0

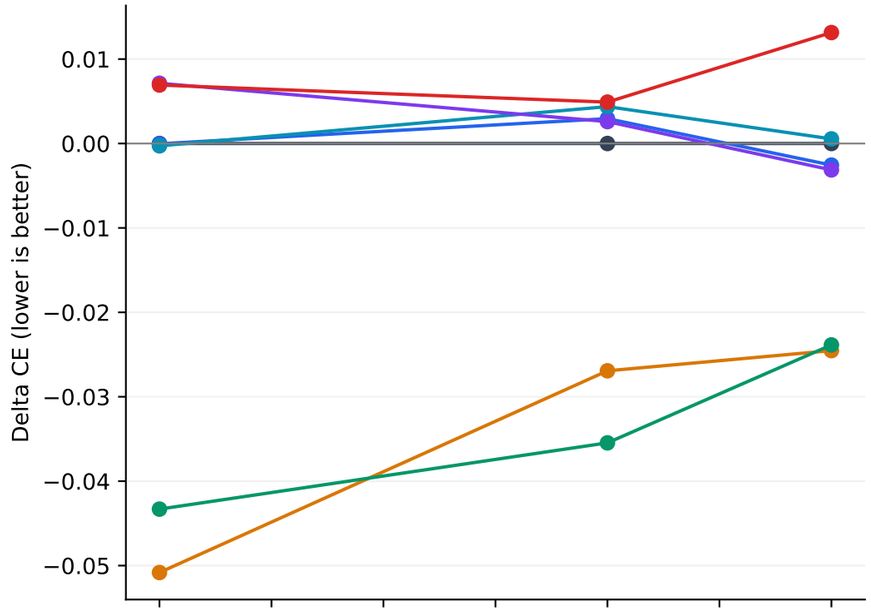
Training loss: trailing 25-step mean



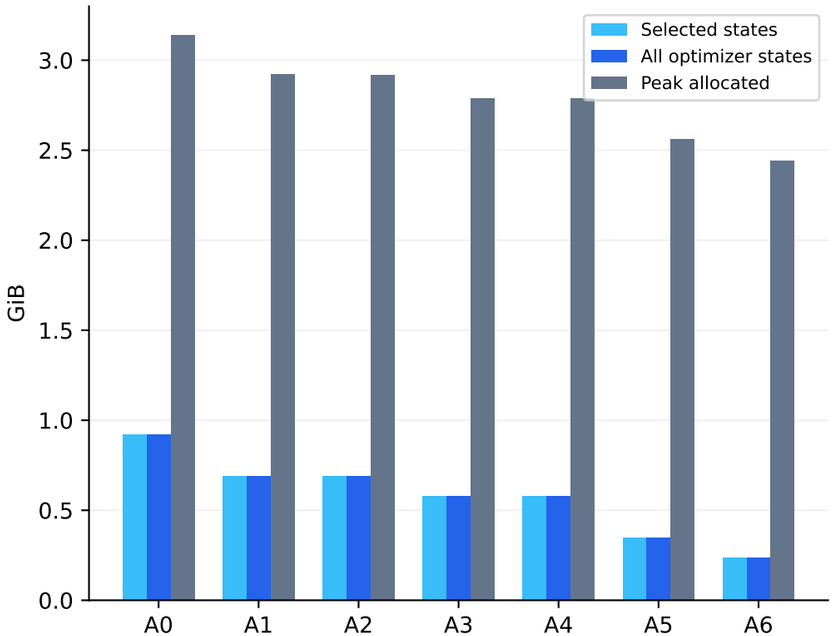
Validation loss: all 3 observations



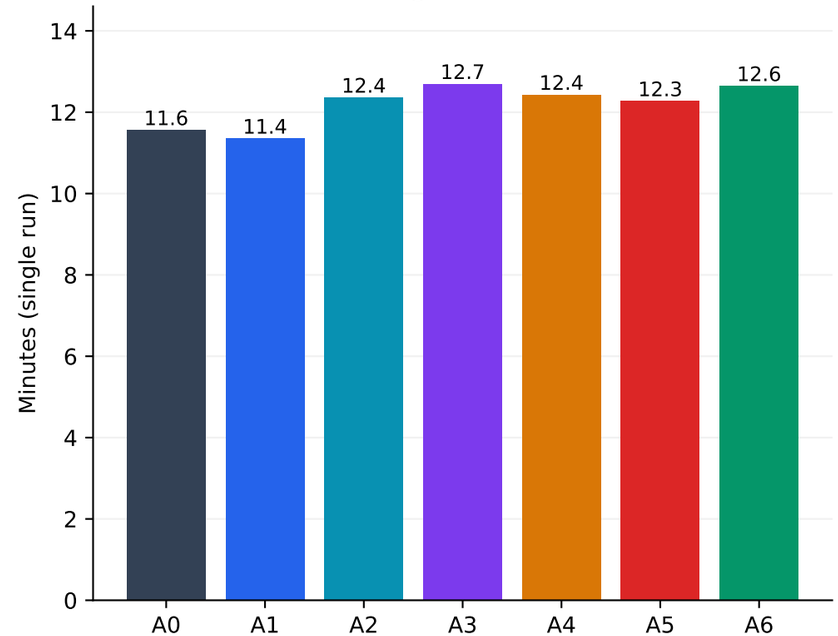
Validation difference vs FP32 baseline



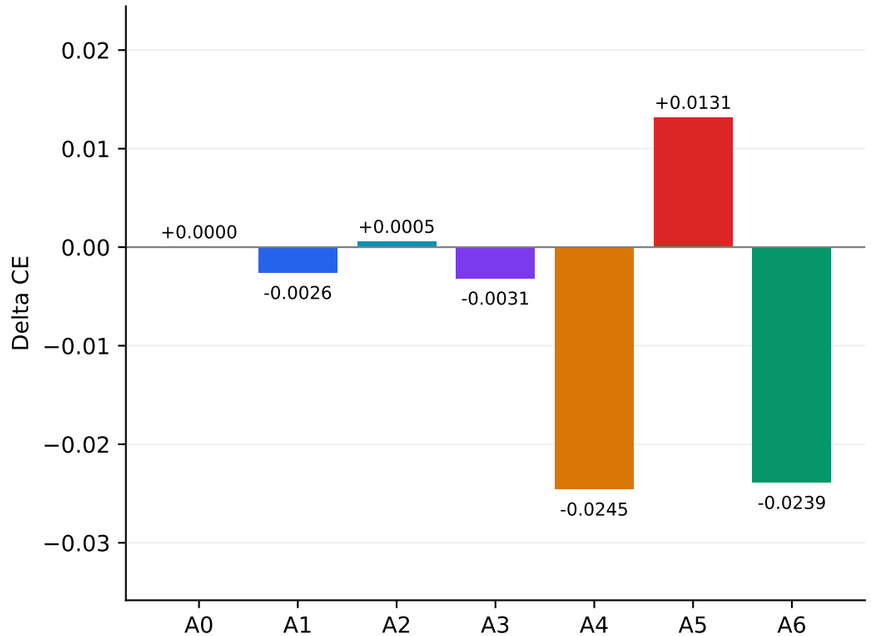
Measured memory (non-additive bars)



Elapsed: training + eval + checkpoints

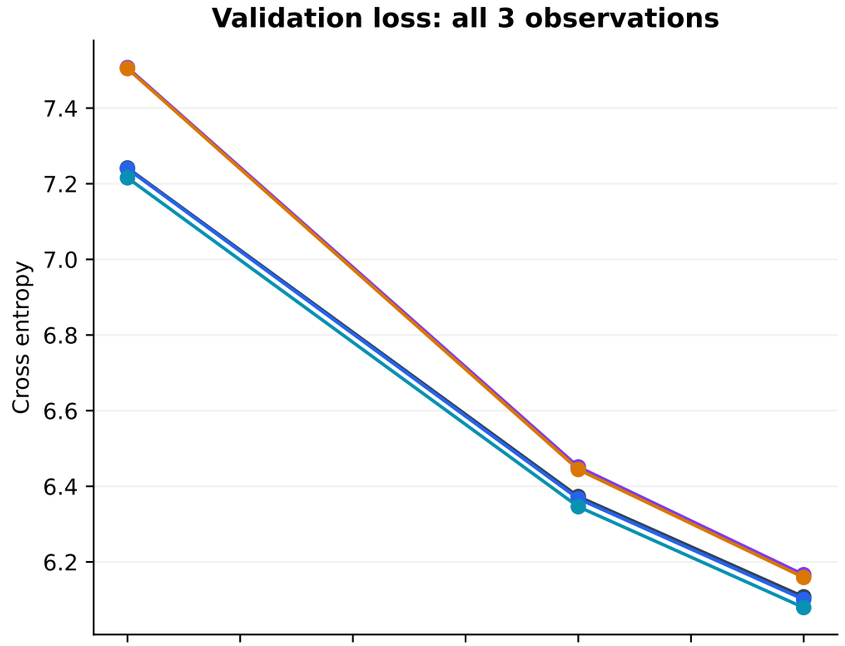


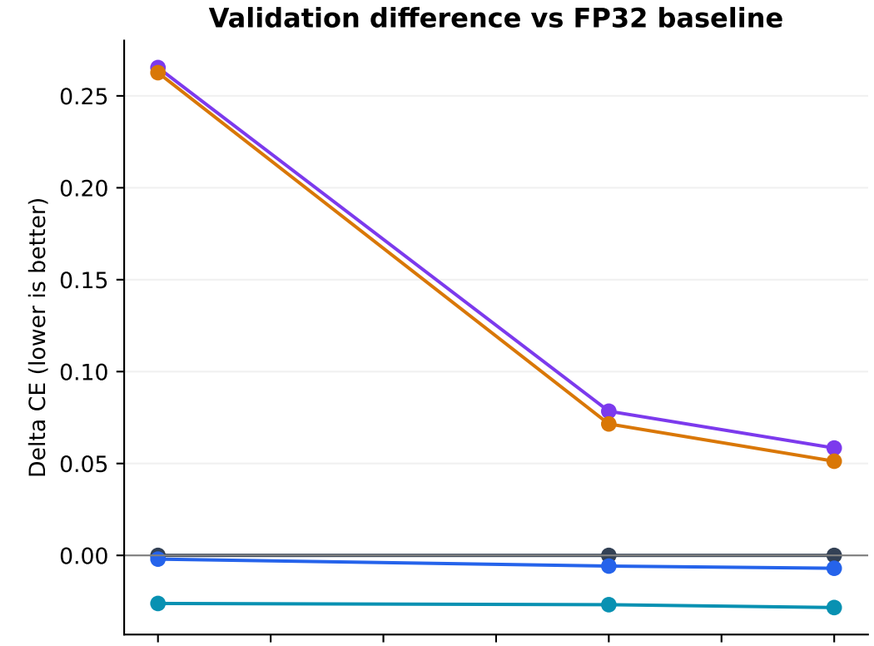
Final validation difference at step 500

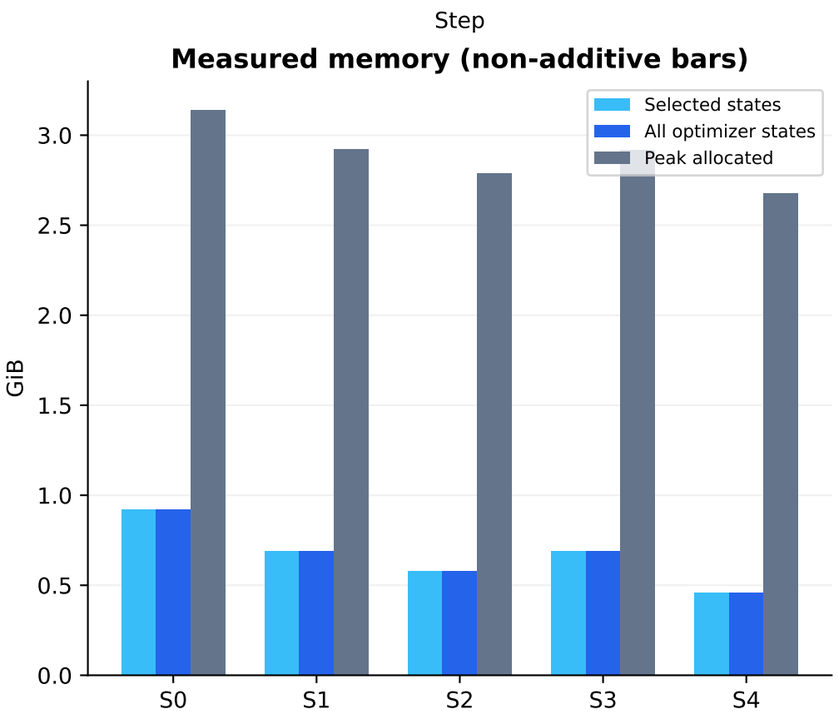


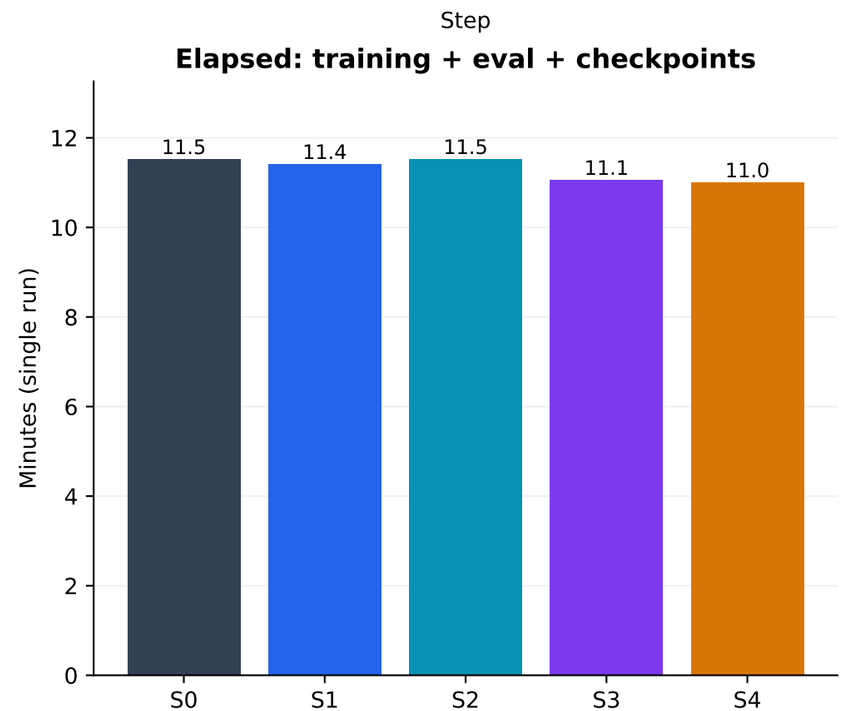
sf-adamw | 124M Llama | packed states | 500 steps | seed 0

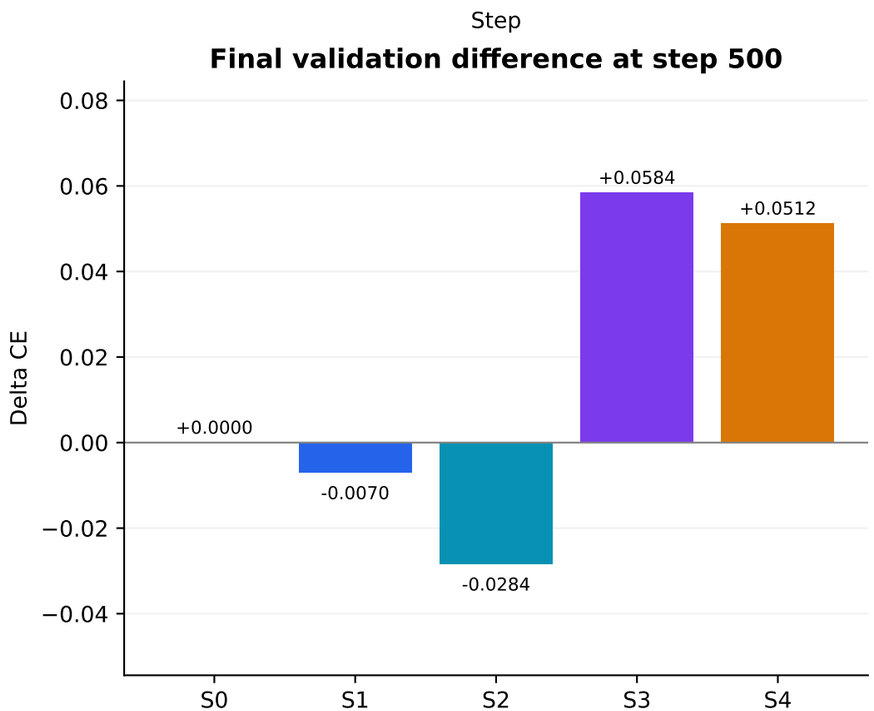






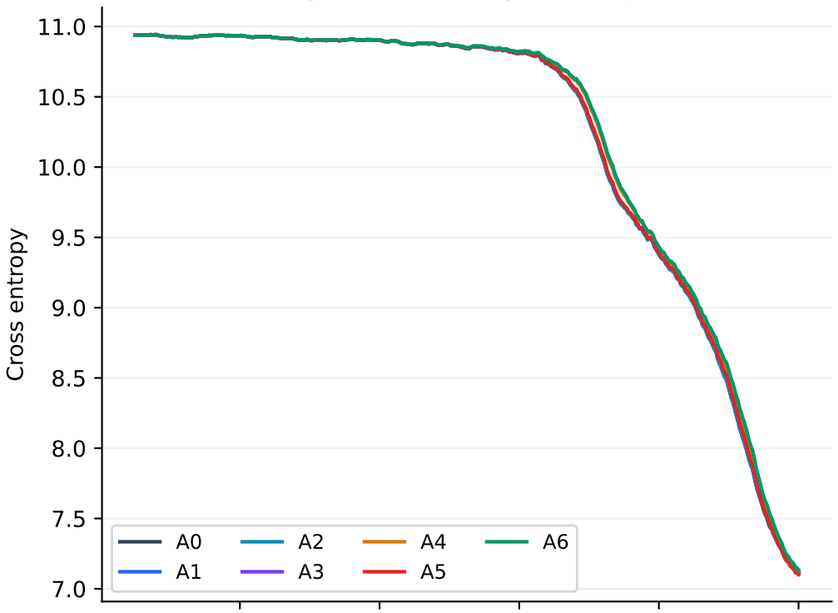




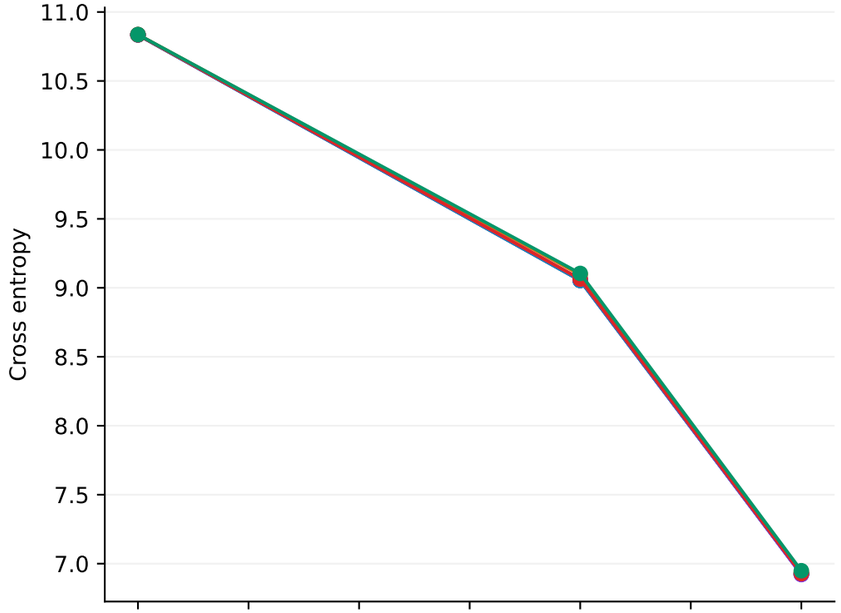


prodigy | 124M Llama | packed states | 500 steps | seed 0

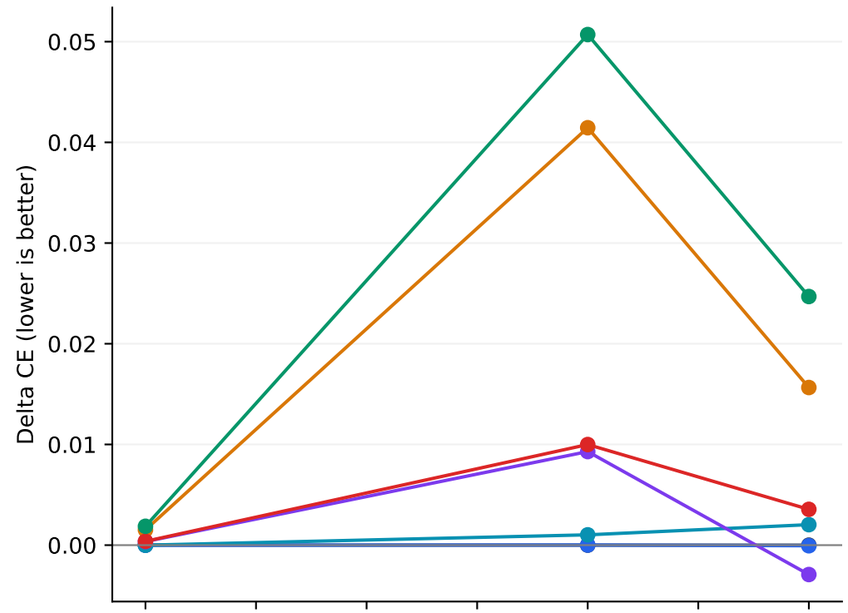
Training loss: trailing 25-step mean



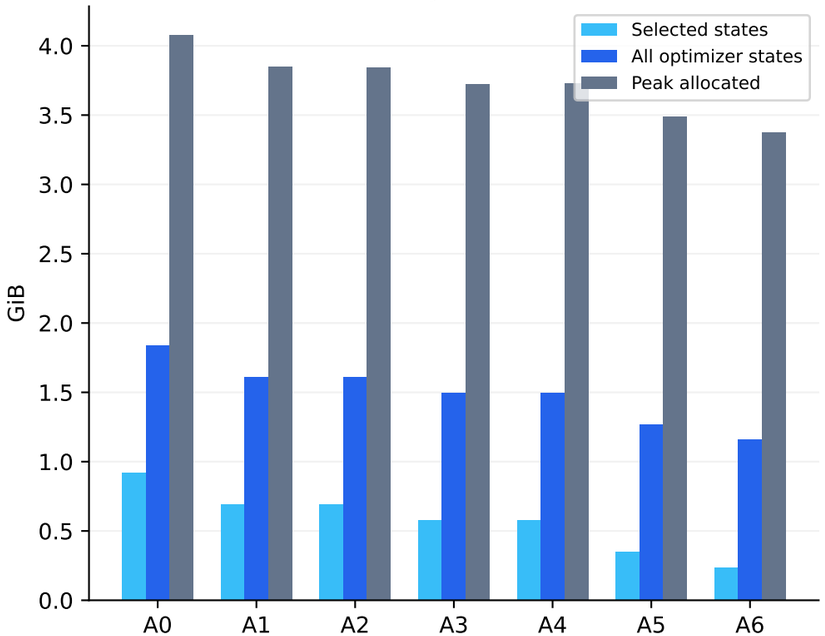
Validation loss: all 3 observations



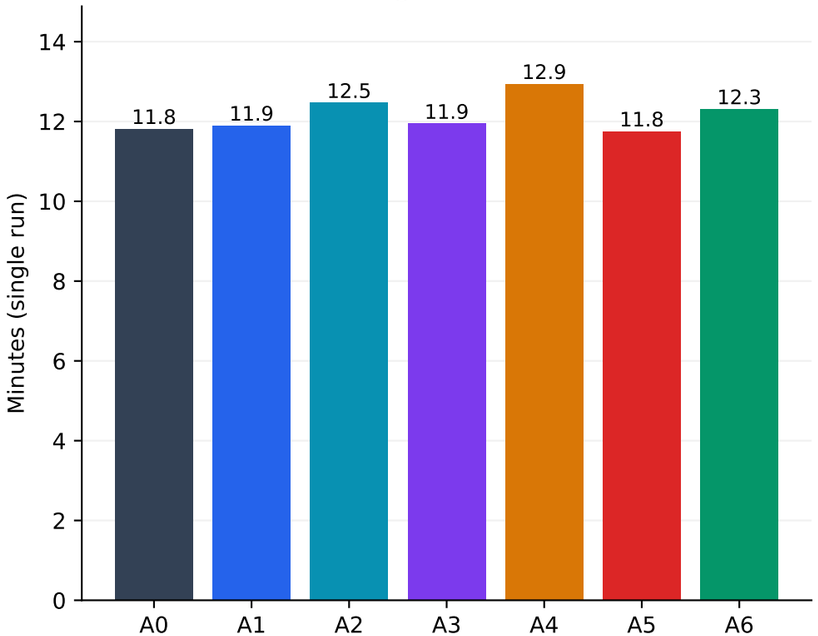
Validation difference vs FP32 baseline



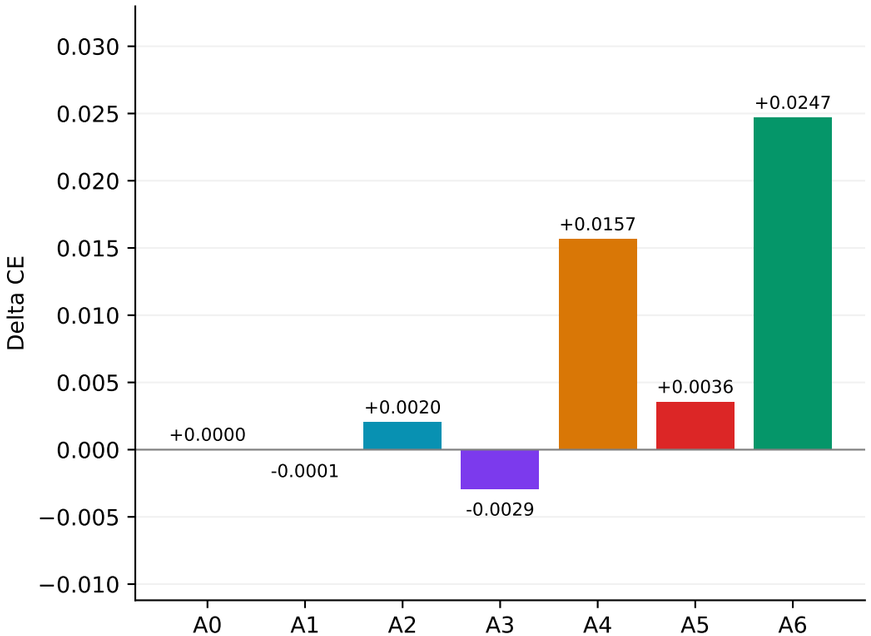
Measured memory (non-additive bars)



Elapsed: training + eval + checkpoints

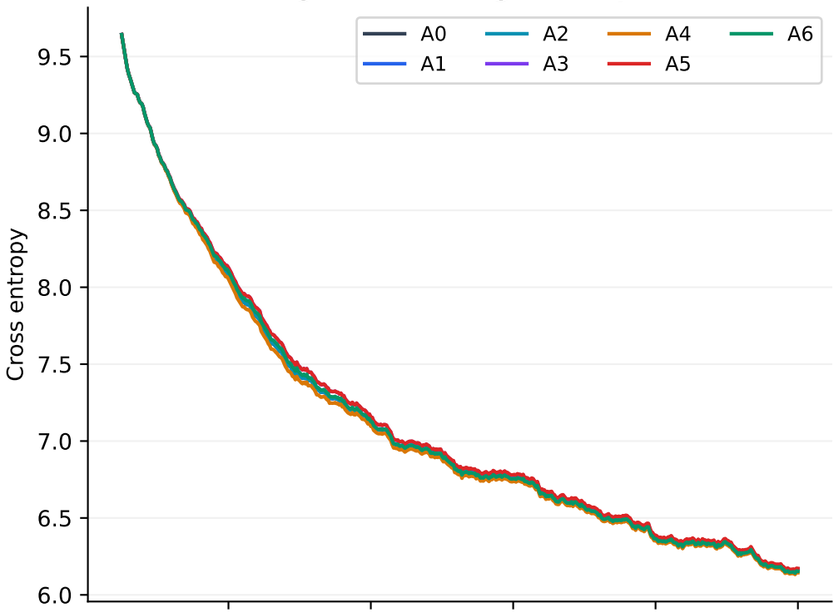


Final validation difference at step 500

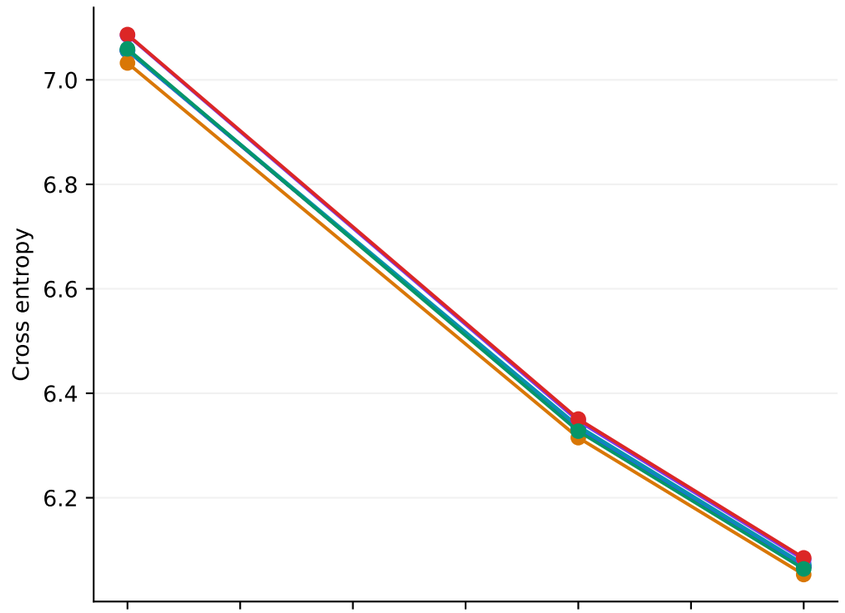


mars-adamw | 124M Llama | packed states | 500 steps | seed 0

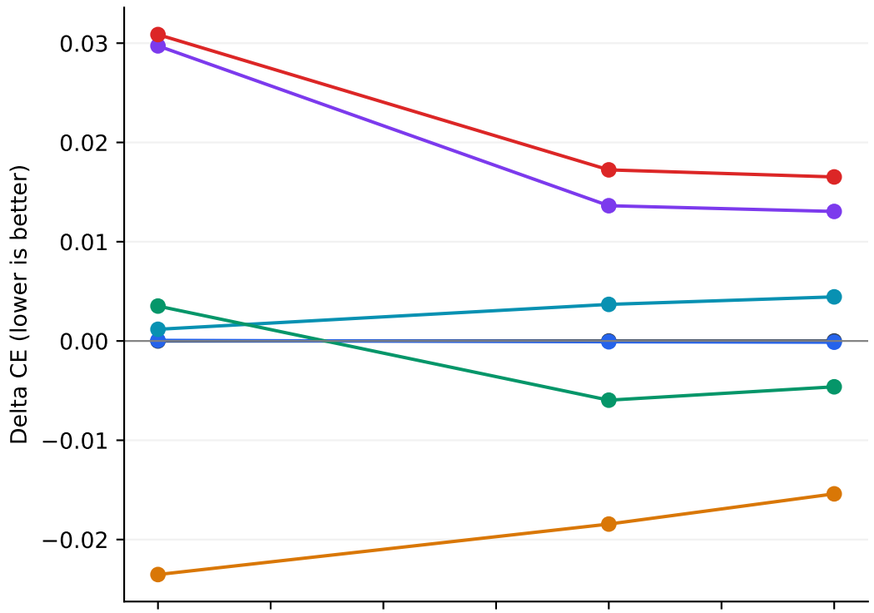
Training loss: trailing 25-step mean



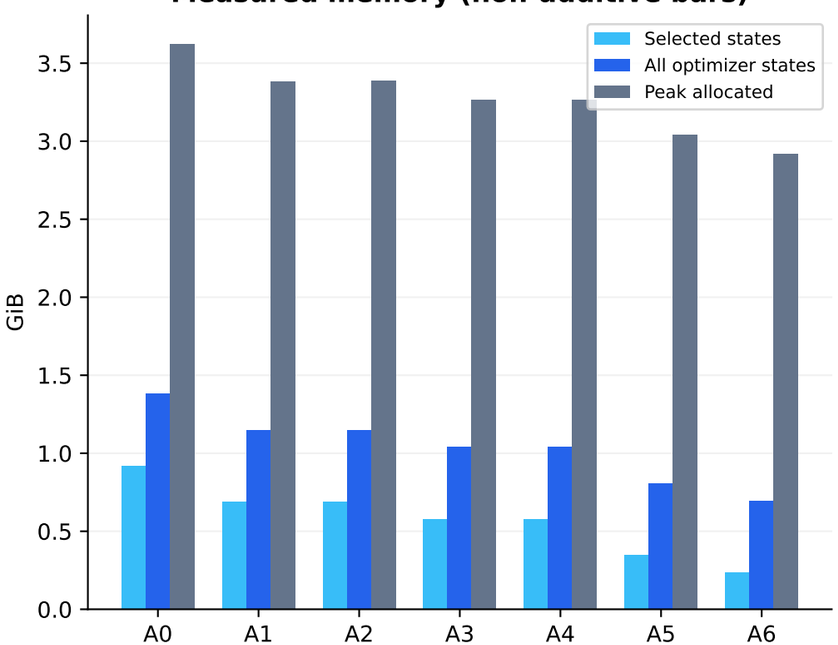
Validation loss: all 3 observations



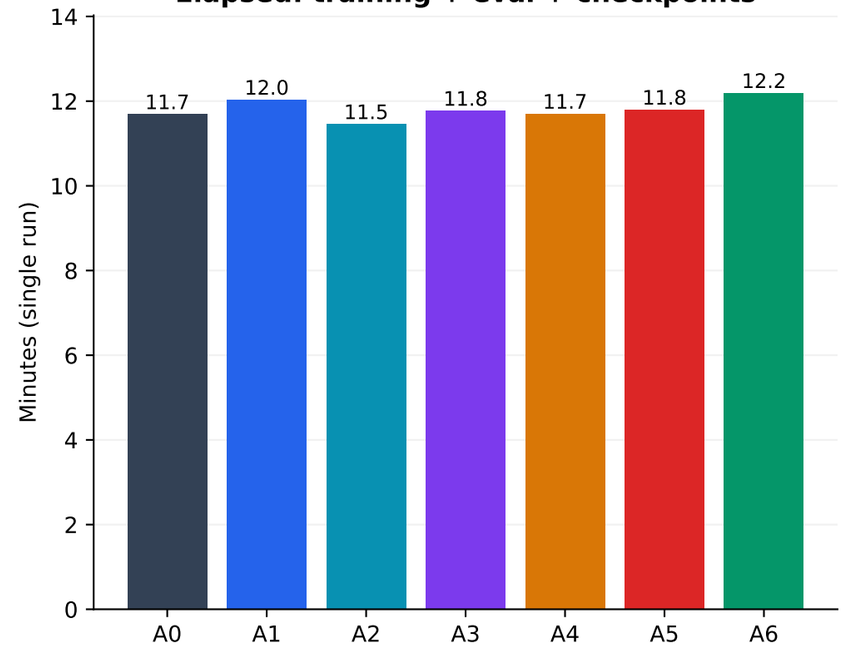
Validation difference vs FP32 baseline



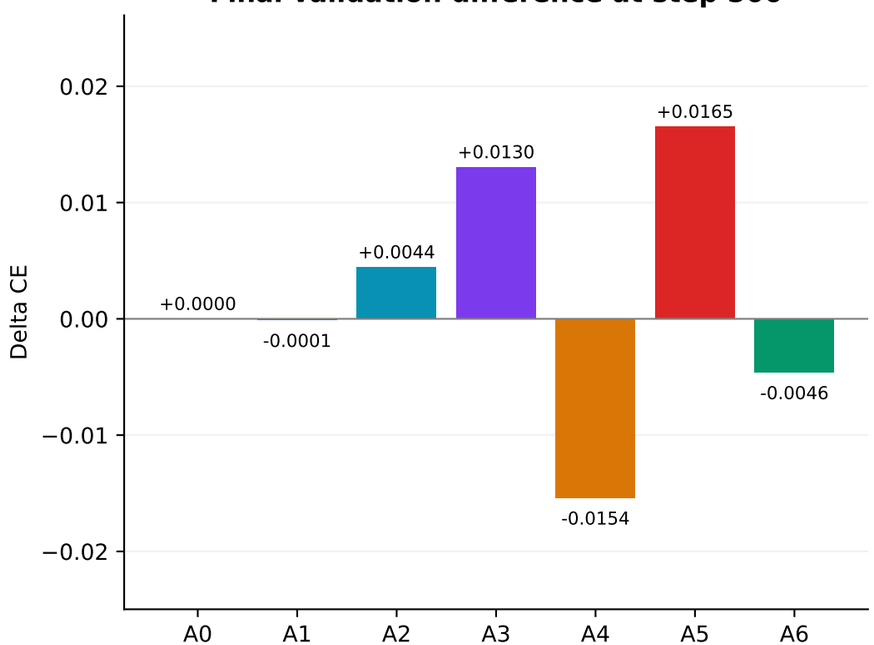
Measured memory (non-additive bars)



Elapsed: training + eval + checkpoints

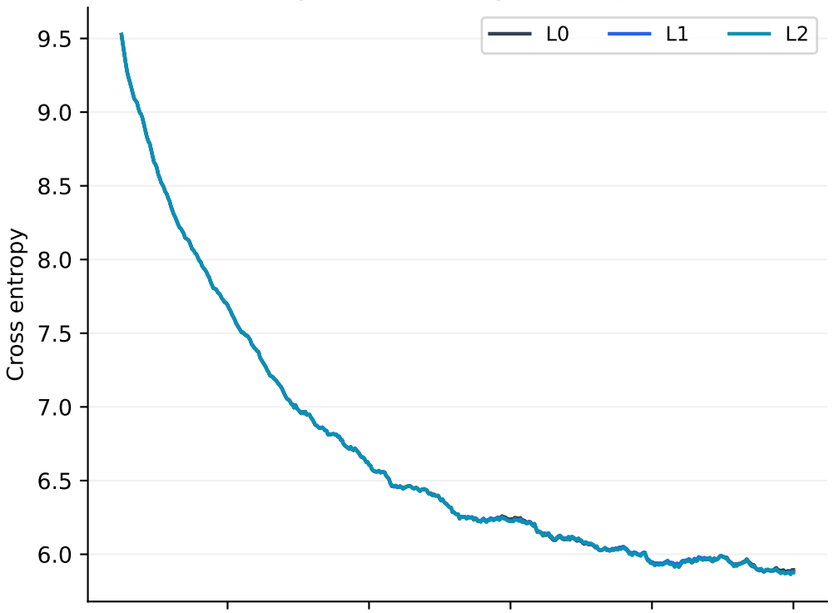


Final validation difference at step 500

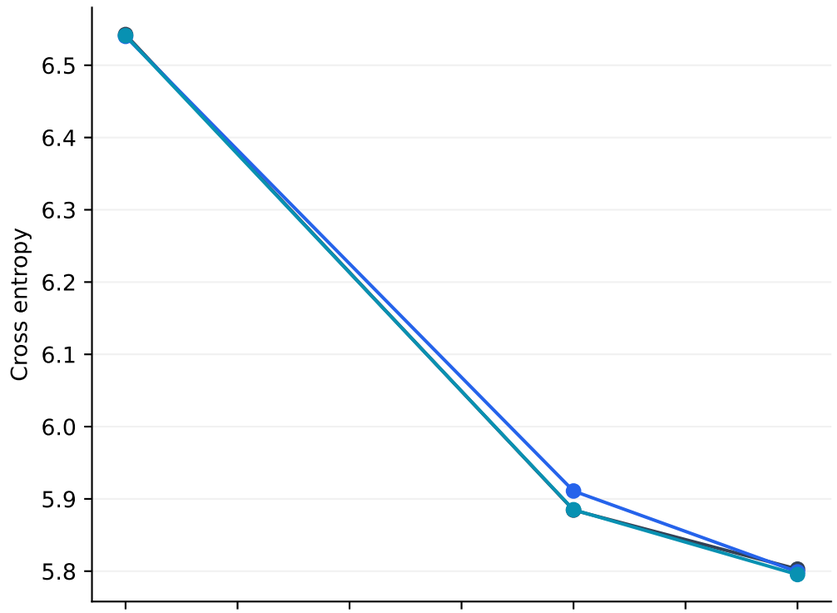


mars-lion | 124M Llama | packed states | 500 steps | seed 0

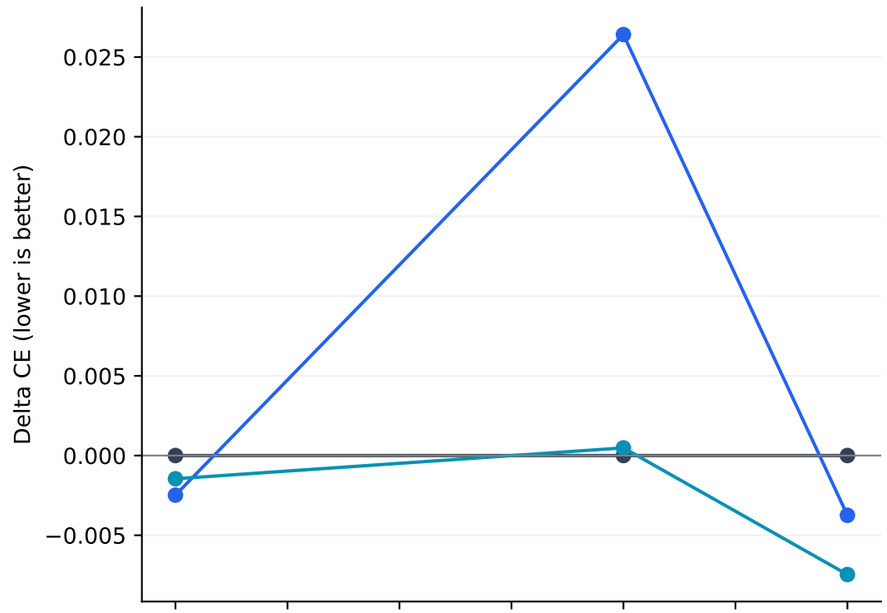
Training loss: trailing 25-step mean



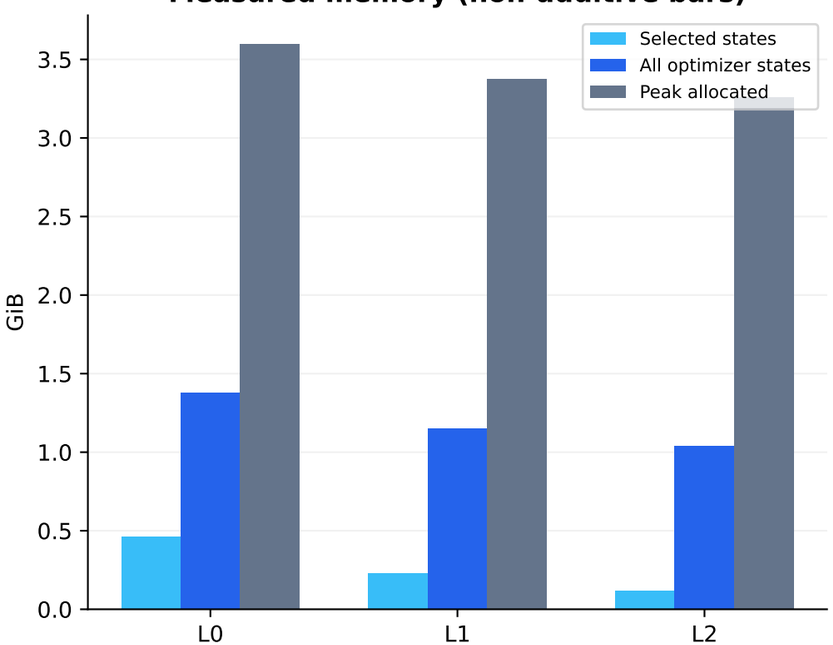
Validation loss: all 3 observations



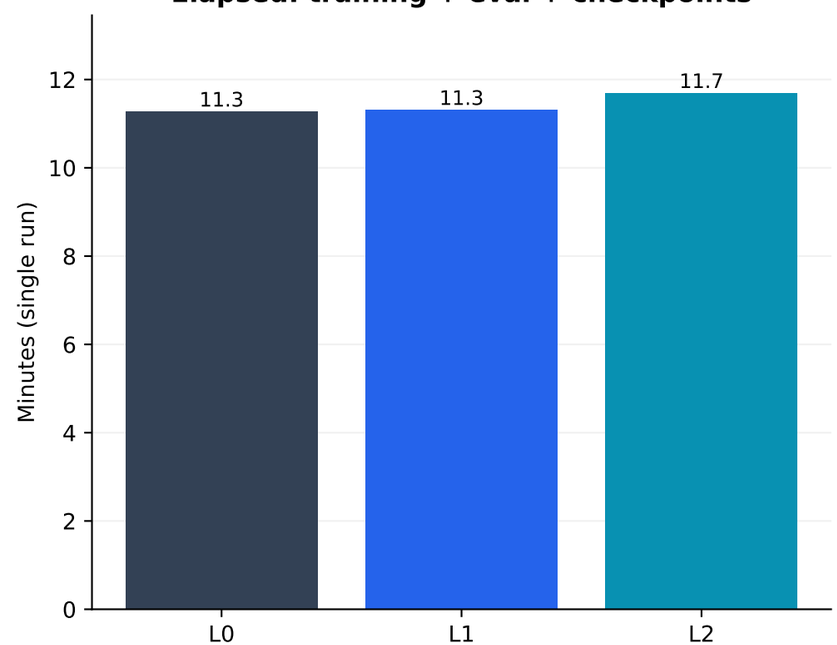
Validation difference vs FP32 baseline



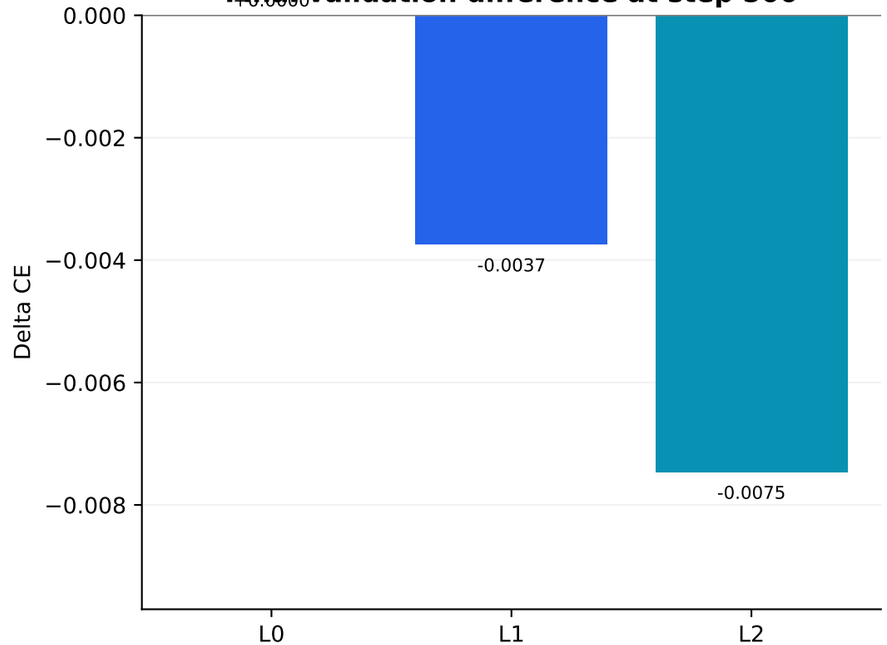
Measured memory (non-additive bars)



Elapsed: training + eval + checkpoints

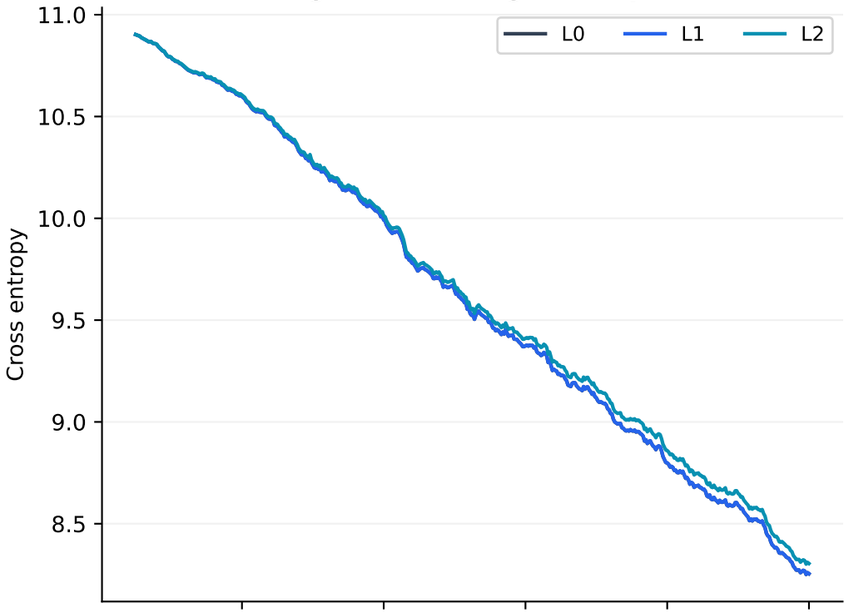


Final validation difference at step 500

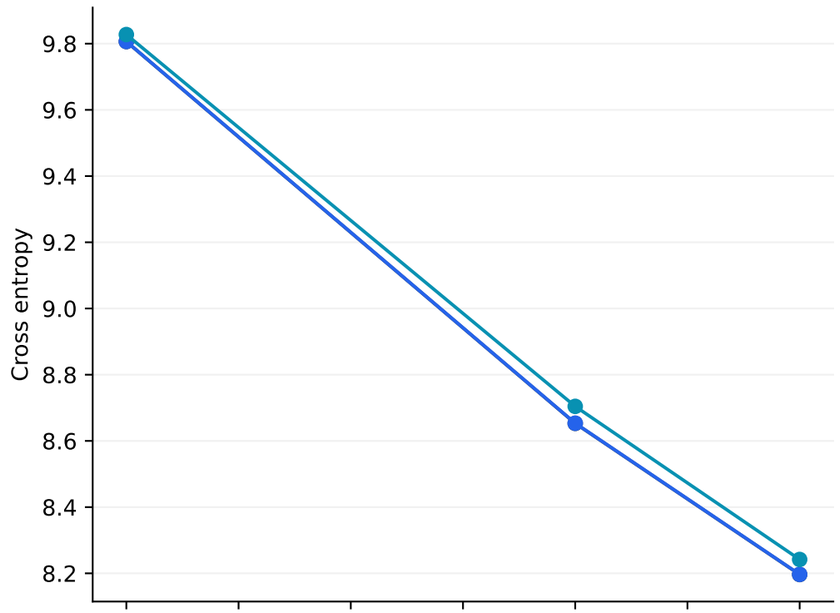


mars-shampoo | 124M Llama | packed states | 500 steps | seed 0

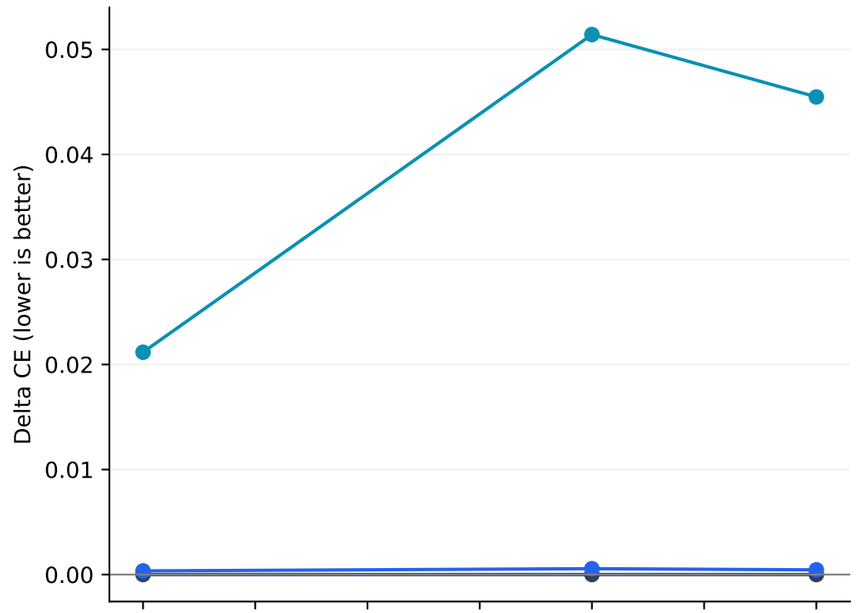
Training loss: trailing 25-step mean



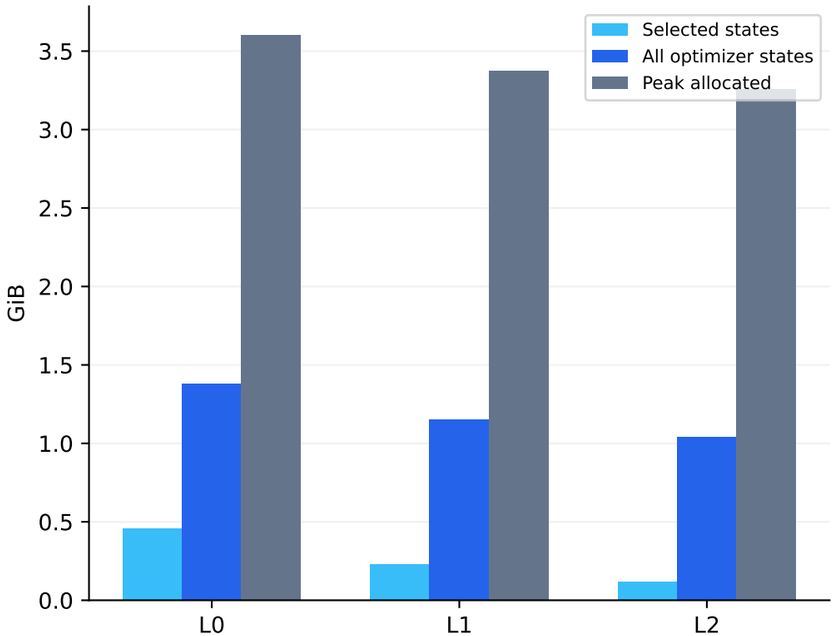
Validation loss: all 3 observations



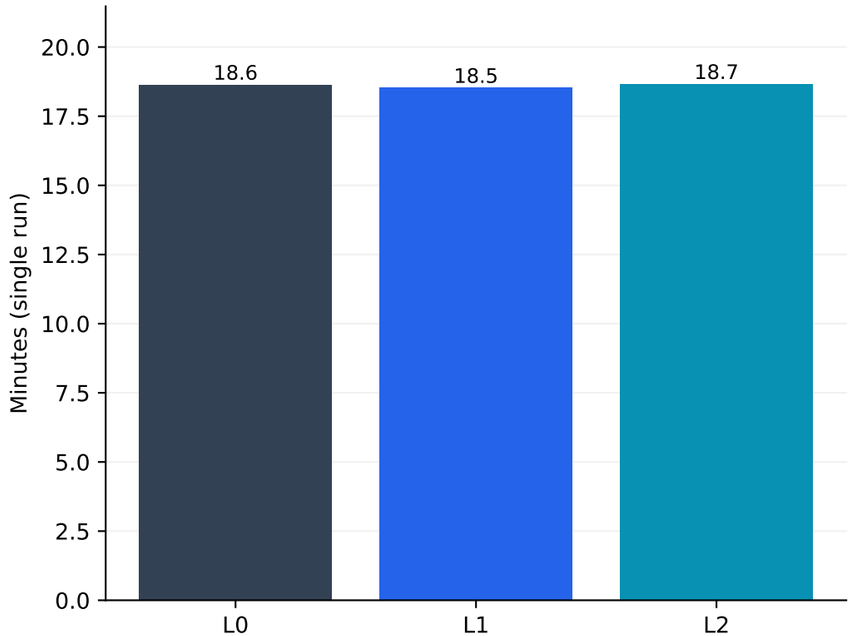
Validation difference vs FP32 baseline



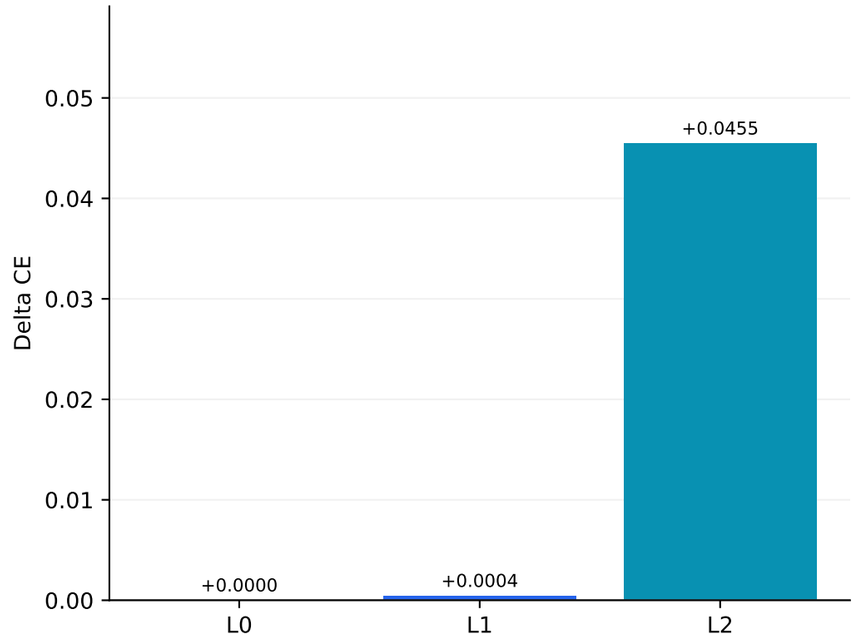
Measured memory (non-additive bars)



Elapsed: training + eval + checkpoints

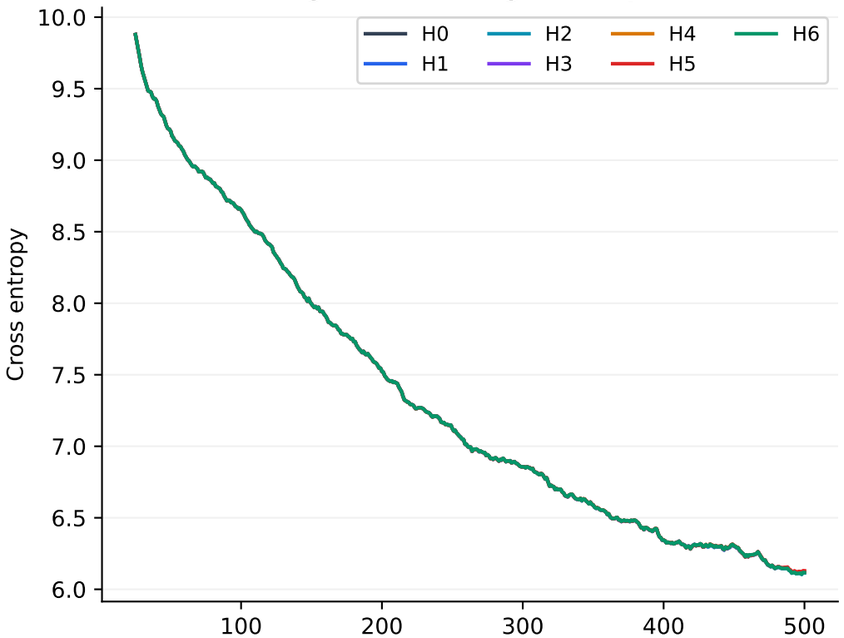


Final validation difference at step 500

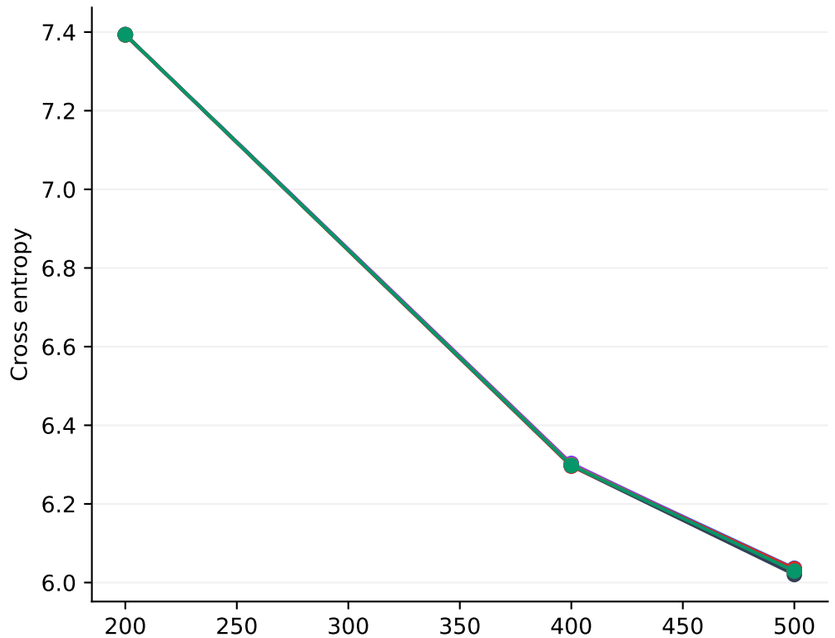


sophiag | 124M Llama | packed states | 500 steps | seed 0

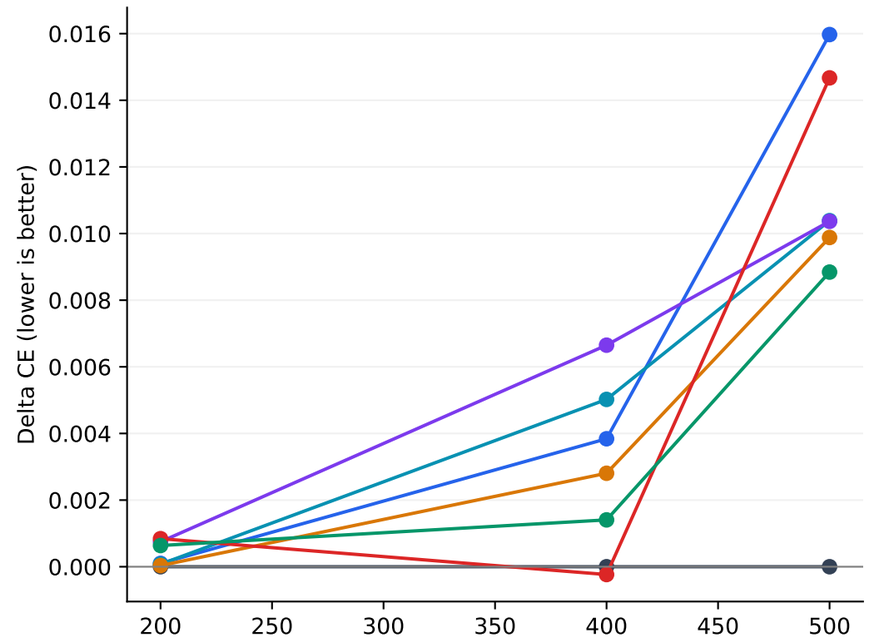
Training loss: trailing 25-step mean



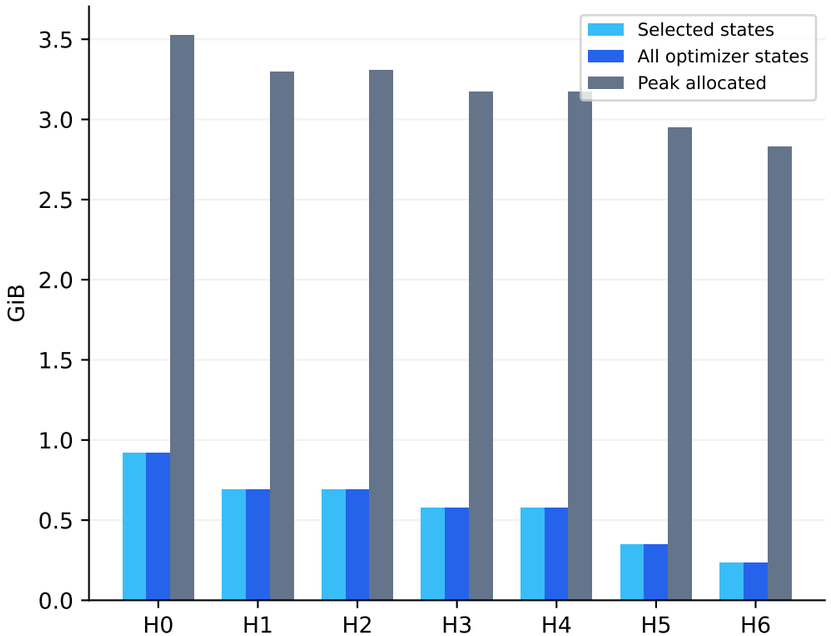
Validation loss: all 3 observations



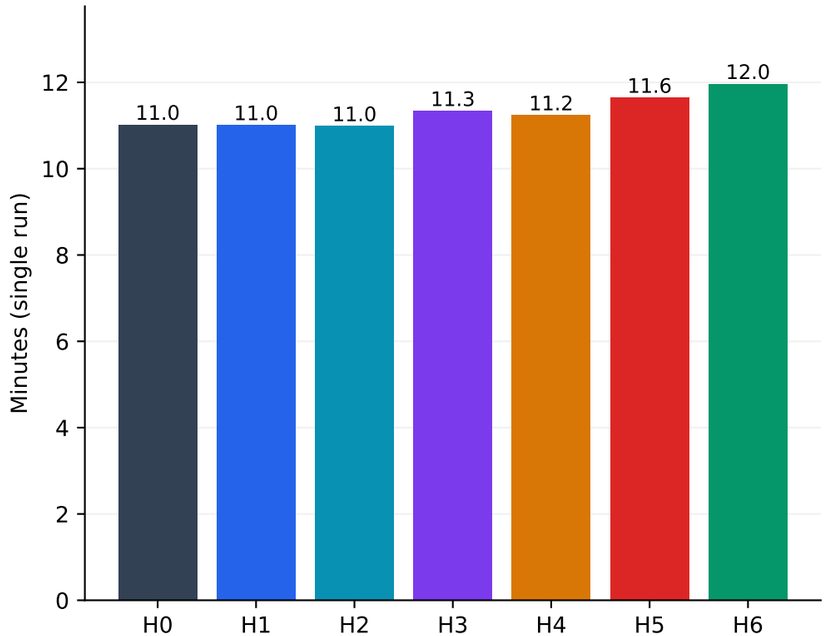
Validation difference vs FP32 baseline



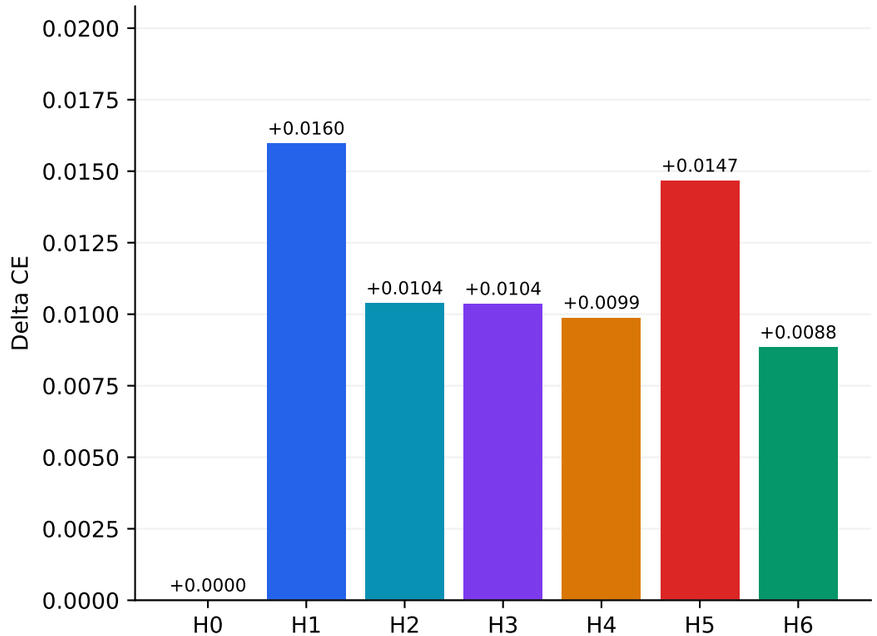
Measured memory (non-additive bars)



Elapsed: training + eval + checkpoints

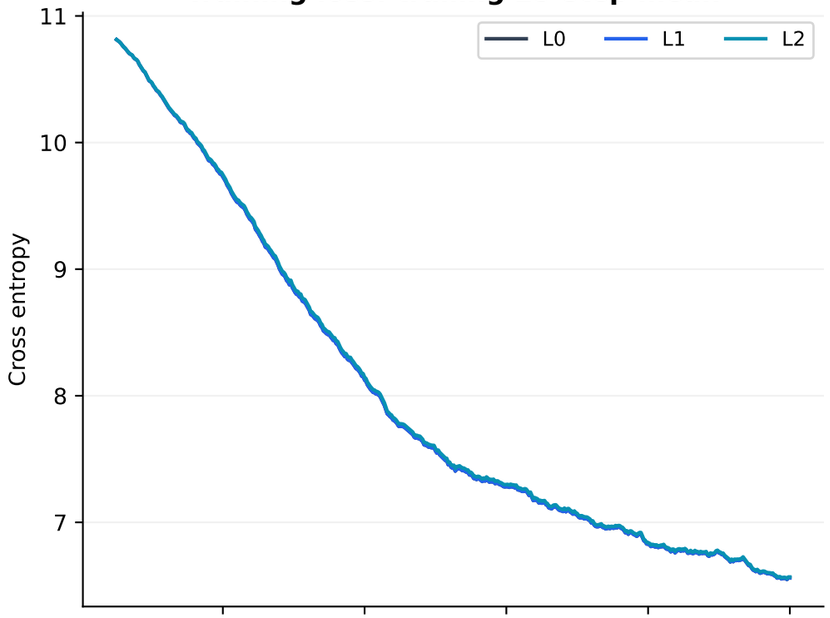


Final validation difference at step 500

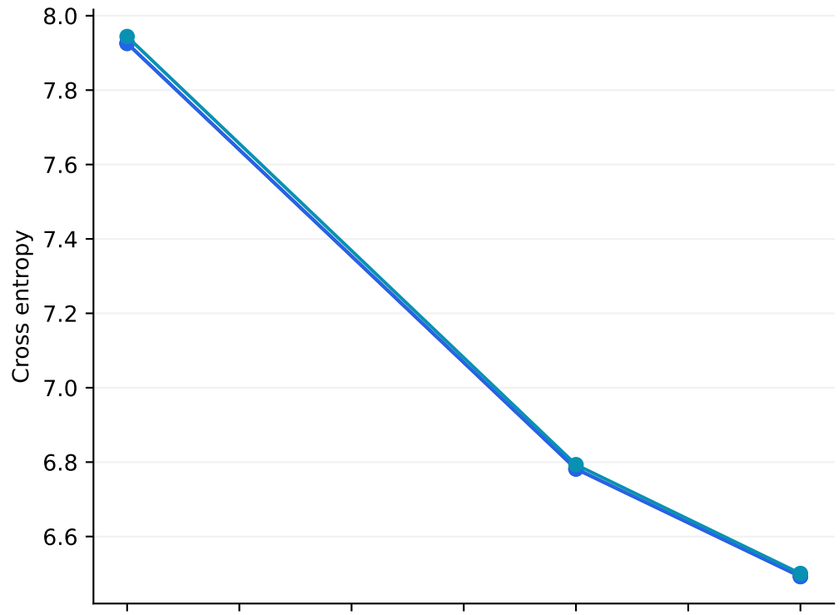


muon | 124M Llama | packed states | 500 steps | seed 0

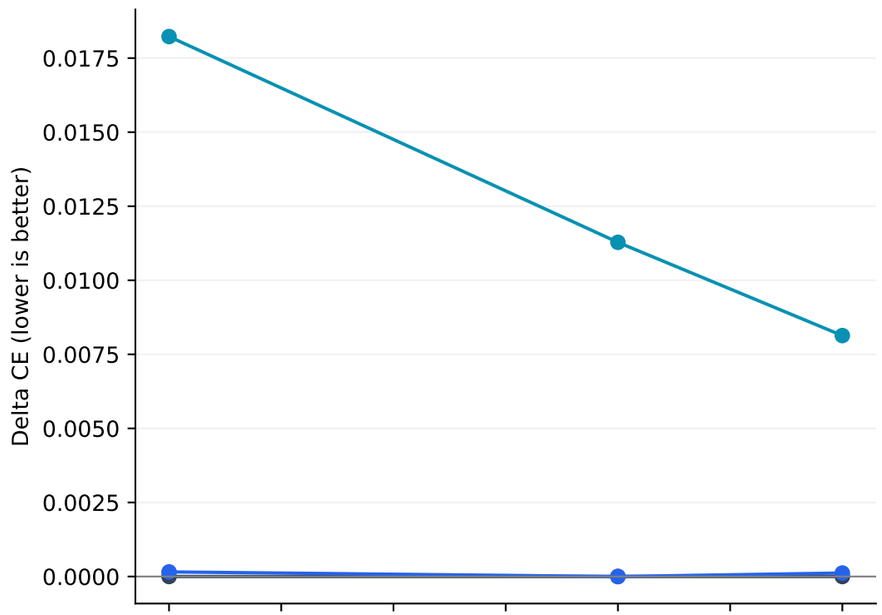
Training loss: trailing 25-step mean



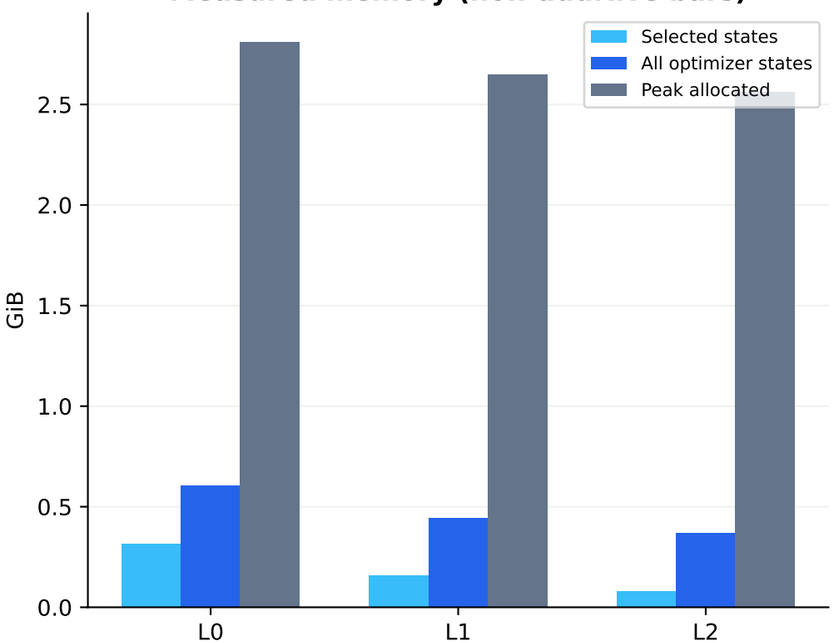
Validation loss: all 3 observations



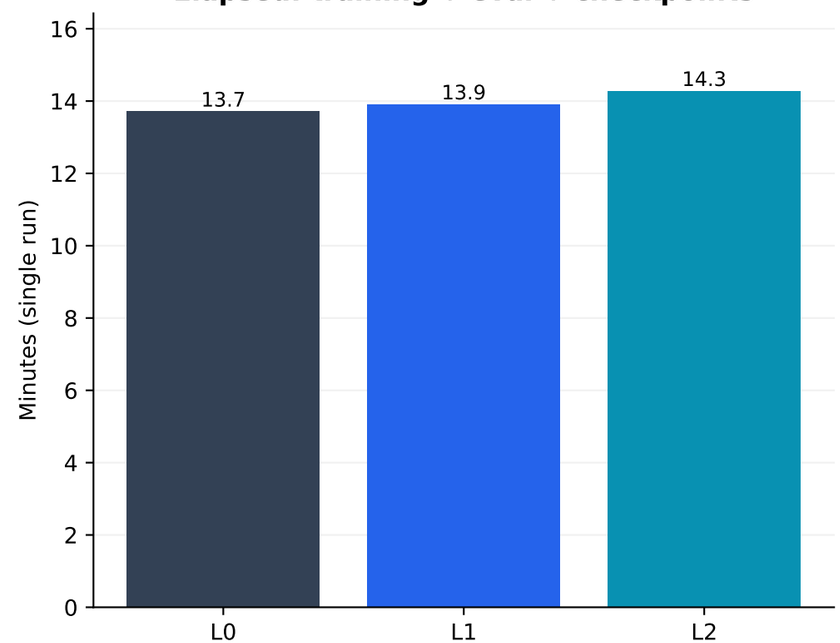
Validation difference vs FP32 baseline



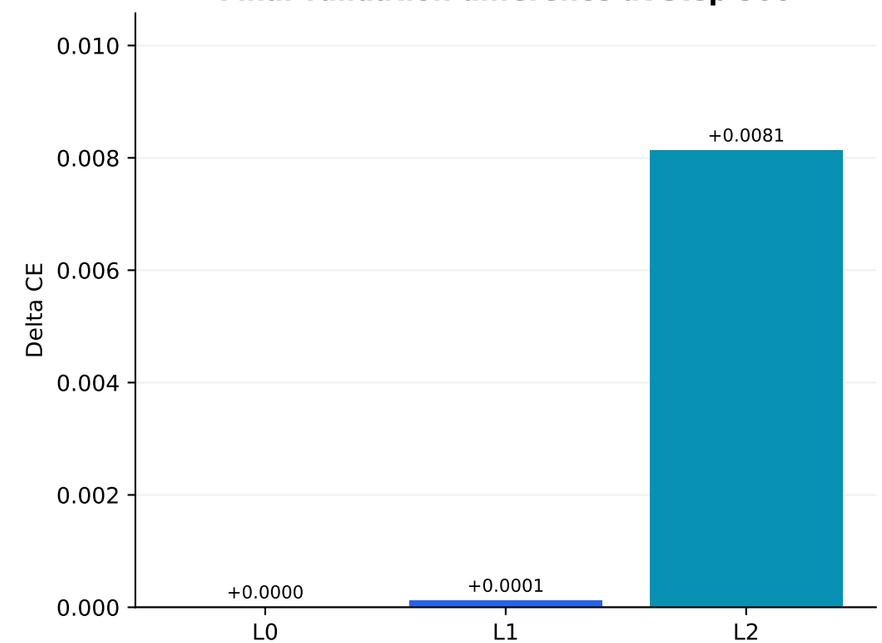
Measured memory (non-additive bars)



Elapsed: training + eval + checkpoints

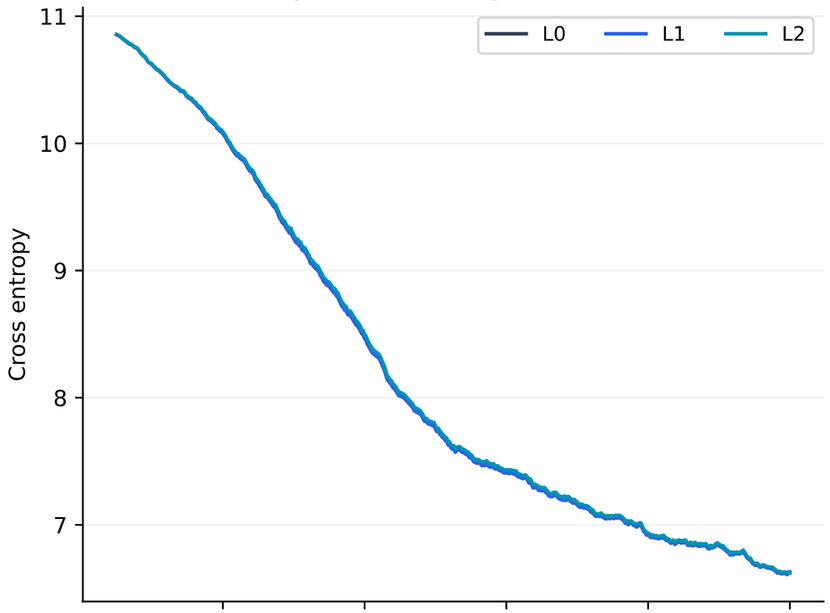


Final validation difference at step 500

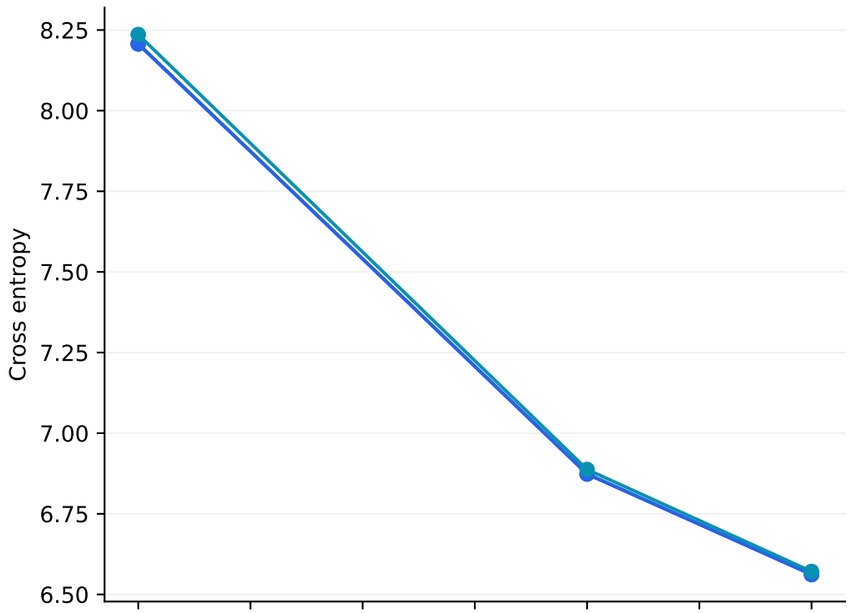


d-muon | 124M Llama | packed states | 500 steps | seed 0

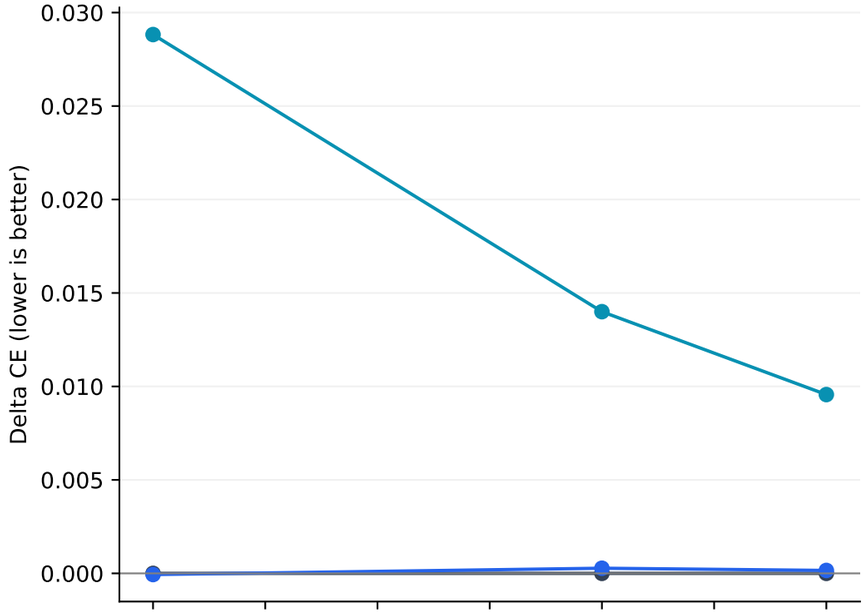
Training loss: trailing 25-step mean



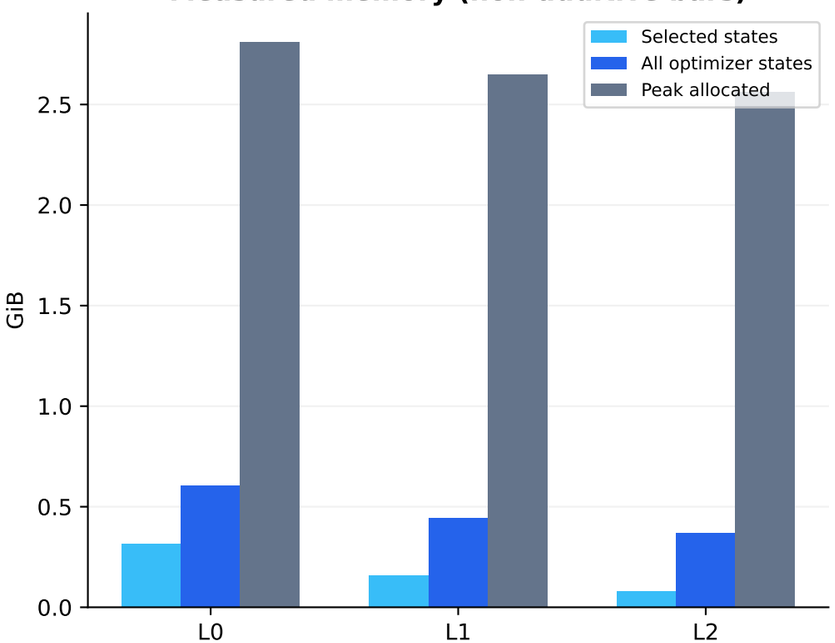
Validation loss: all 3 observations



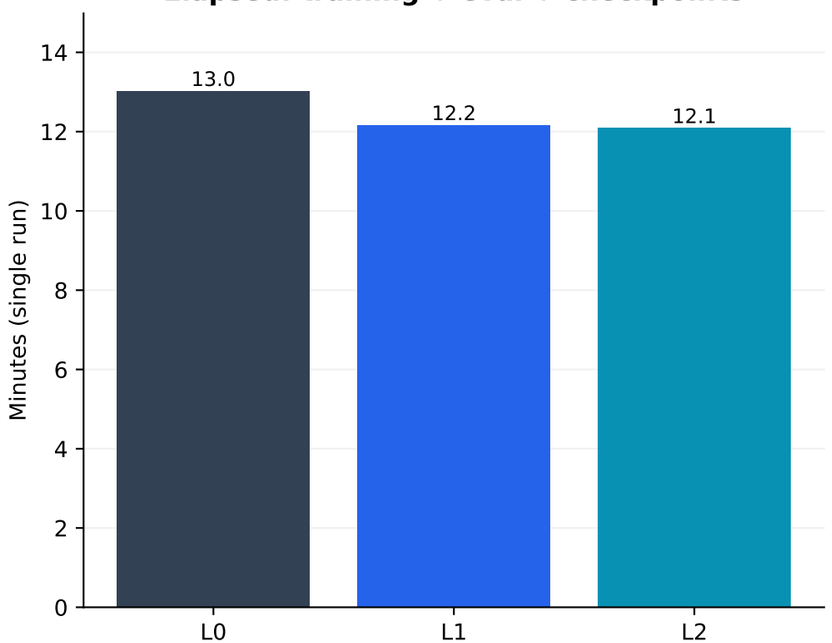
Validation difference vs FP32 baseline



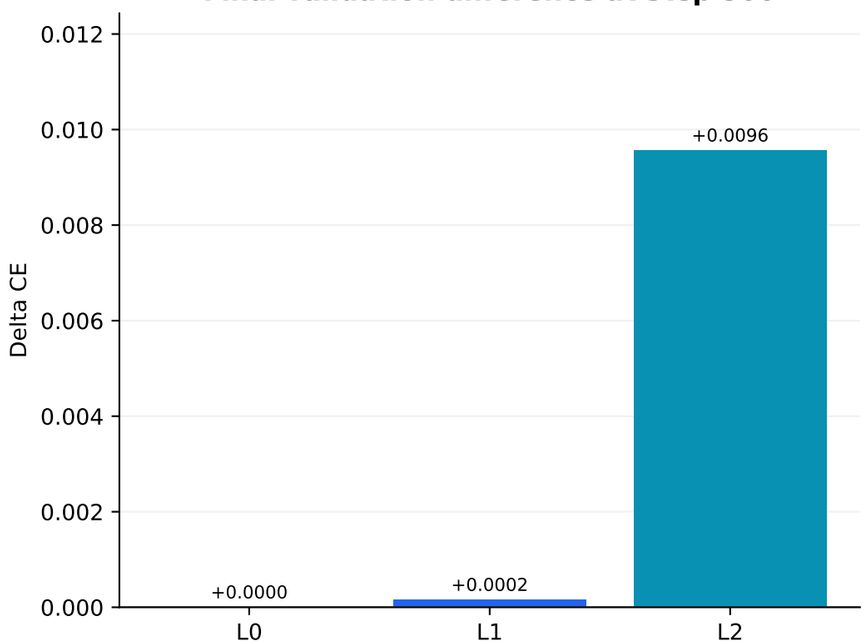
Measured memory (non-additive bars)



Elapsed: training + eval + checkpoints

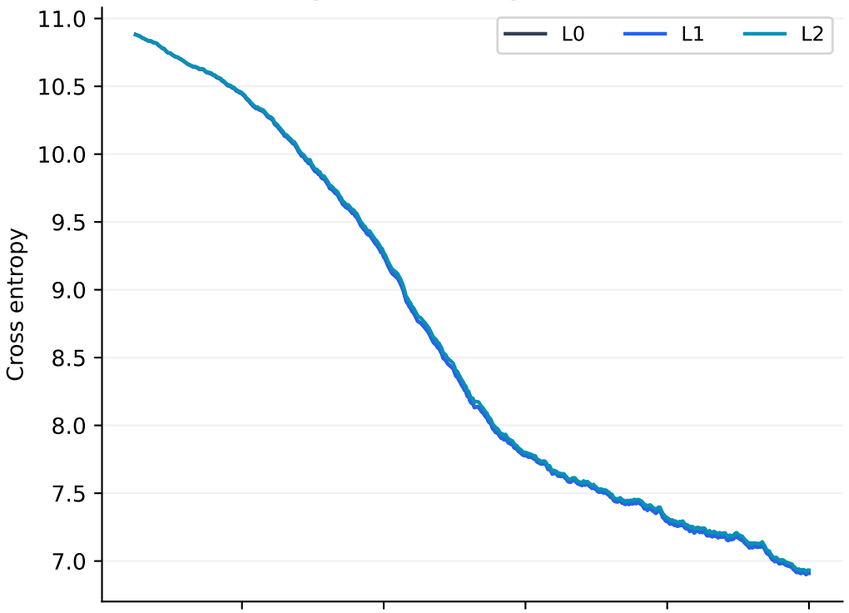


Final validation difference at step 500

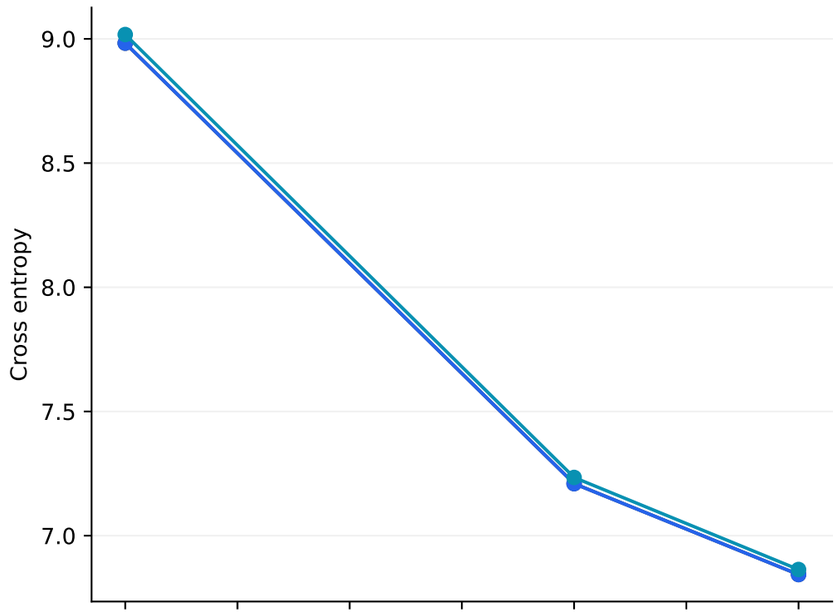


muon-magma | 124M Llama | packed states | 500 steps | seed 0

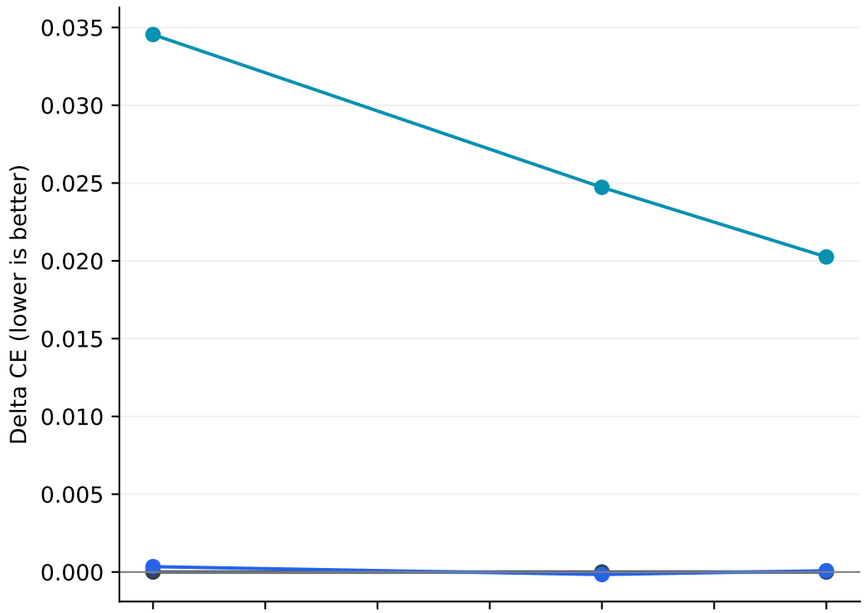
Training loss: trailing 25-step mean



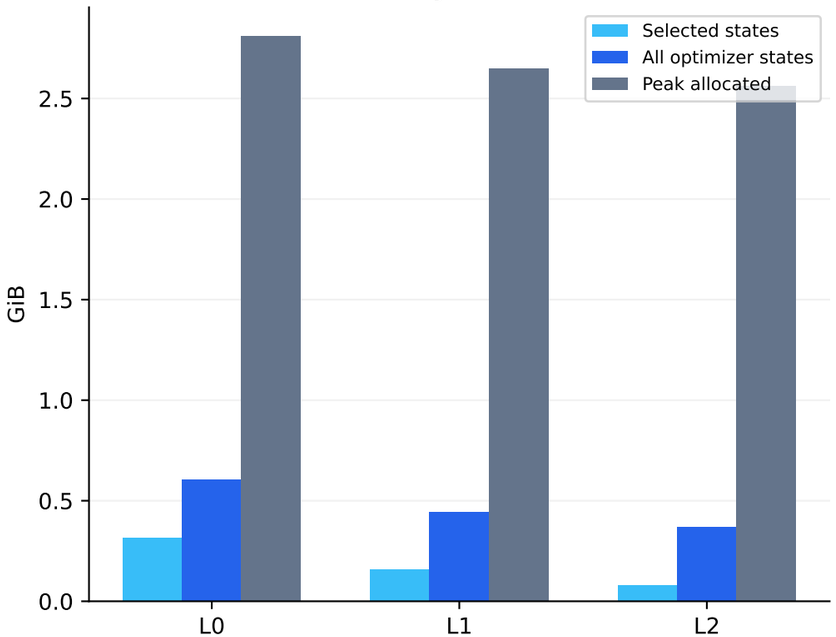
Validation loss: all 3 observations



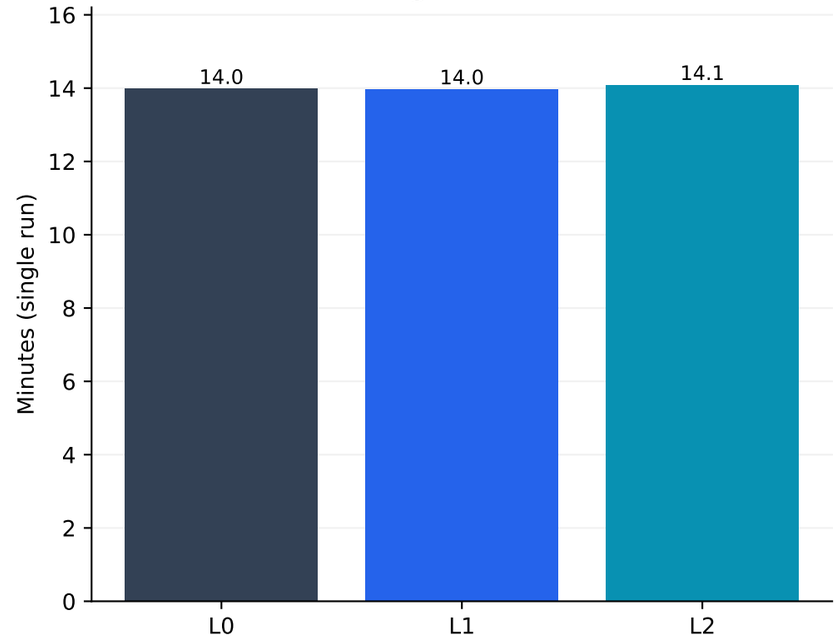
Validation difference vs FP32 baseline



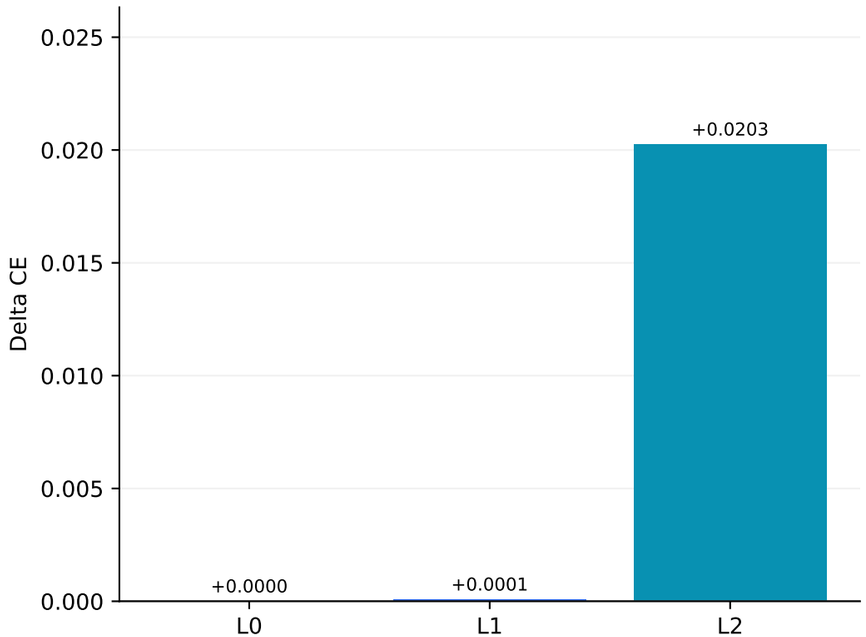
Measured memory (non-additive bars)



Elapsed: training + eval + checkpoints

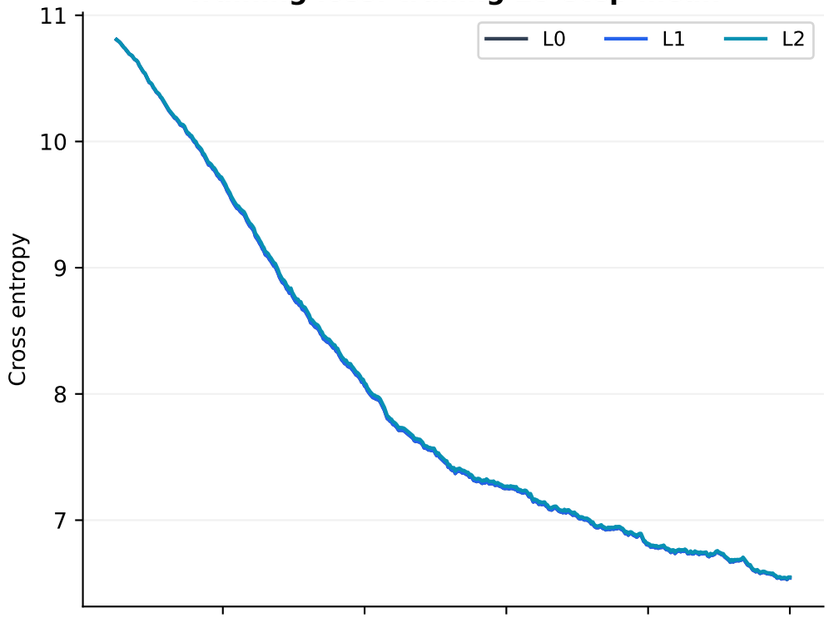


Final validation difference at step 500

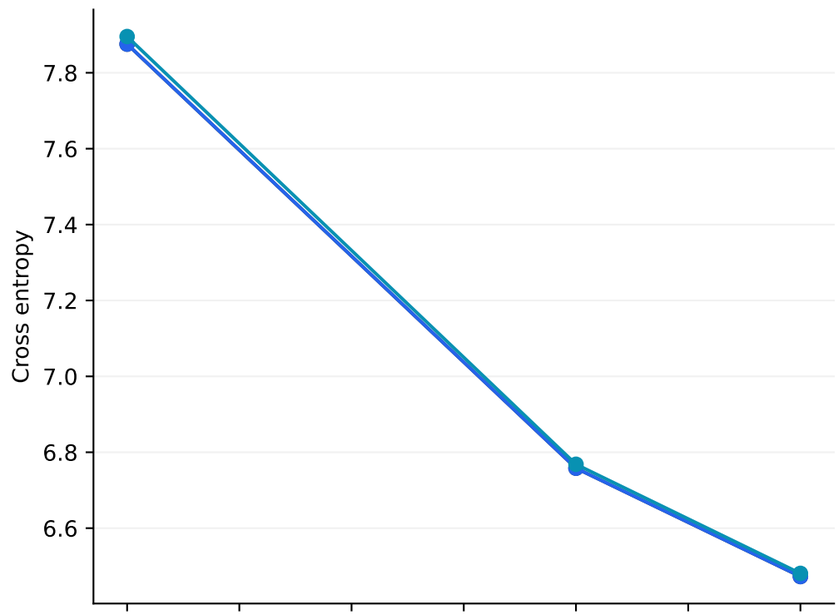


softeq-k2000-muon | 124M Llama | packed states | 500 steps | seed 0

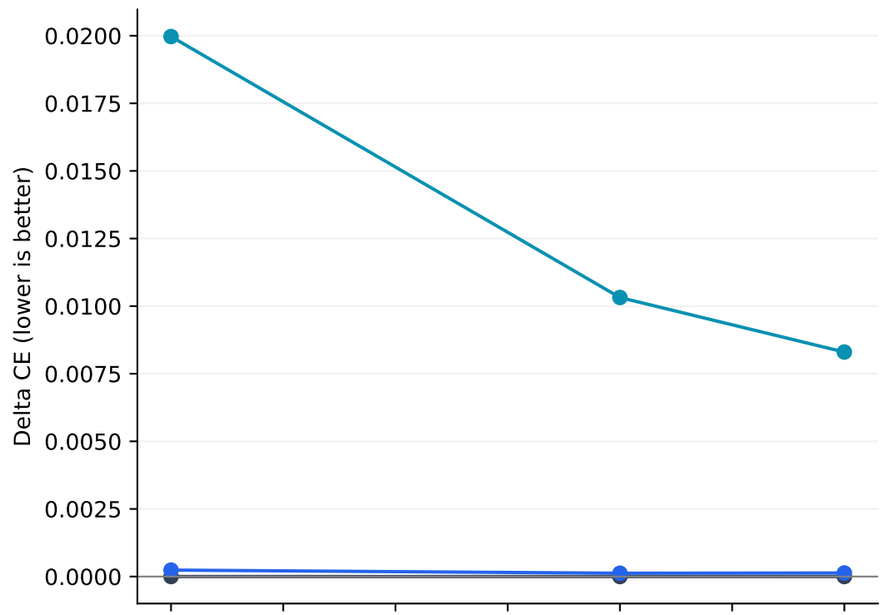
Training loss: trailing 25-step mean



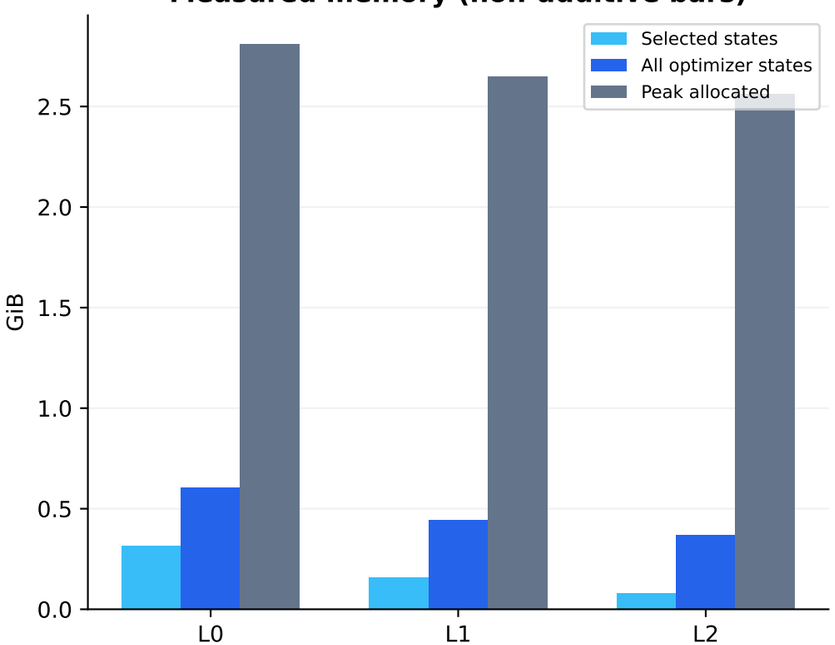
Validation loss: all 3 observations



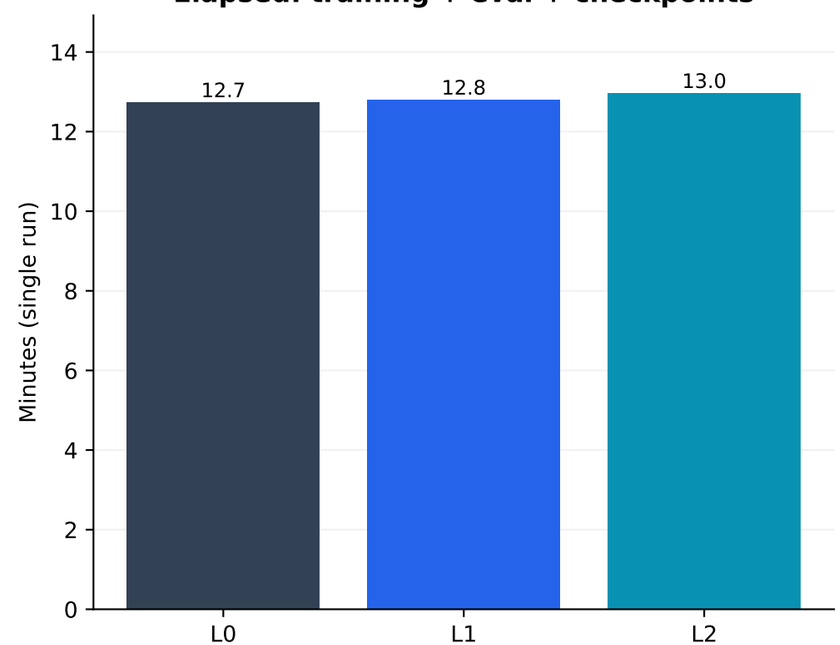
Validation difference vs FP32 baseline



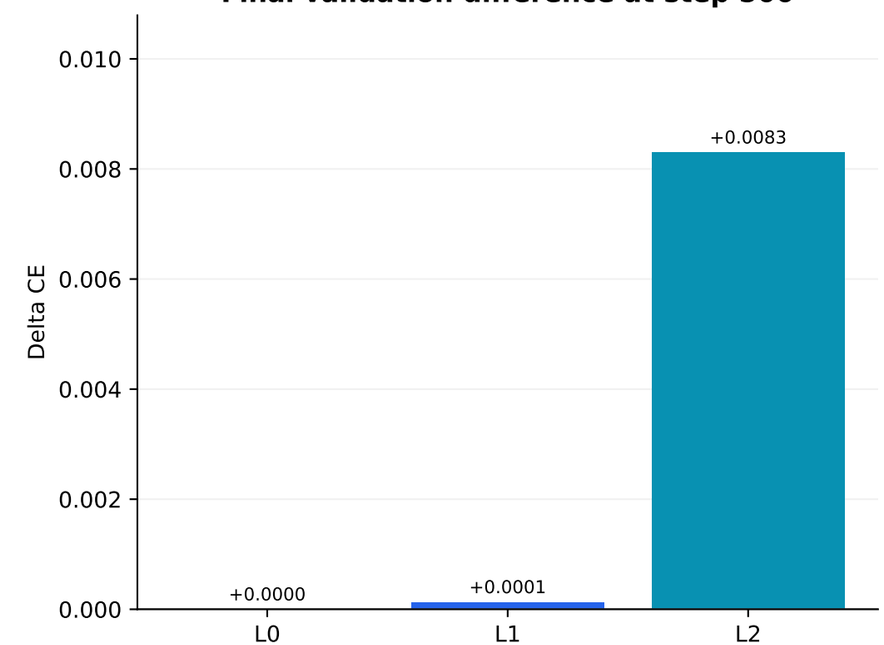
Measured memory (non-additive bars)



Elapsed: training + eval + checkpoints

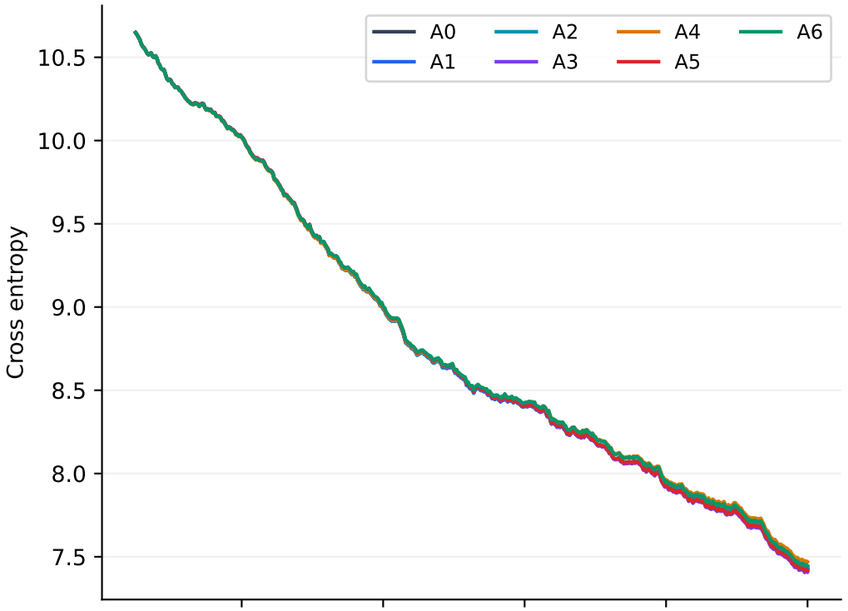


Final validation difference at step 500

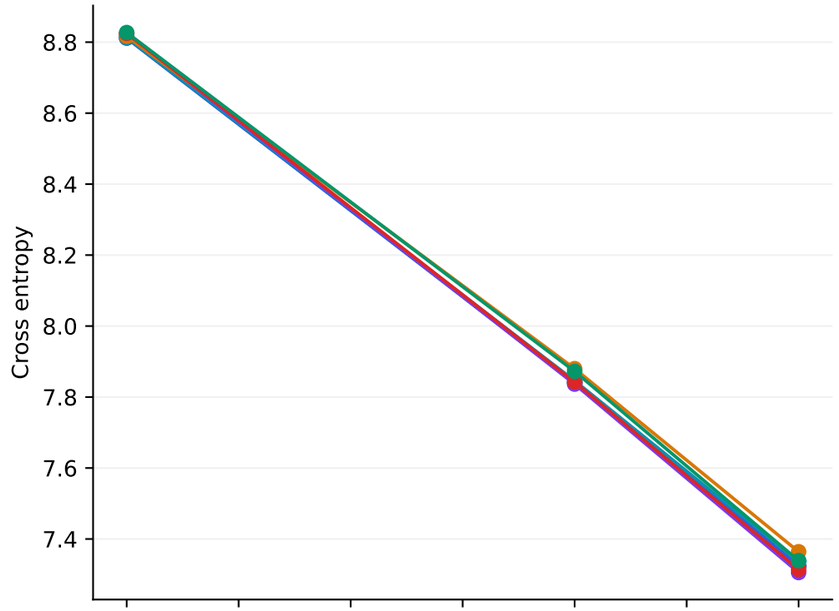


soap | 124M Llama | packed states | 500 steps | seed 0

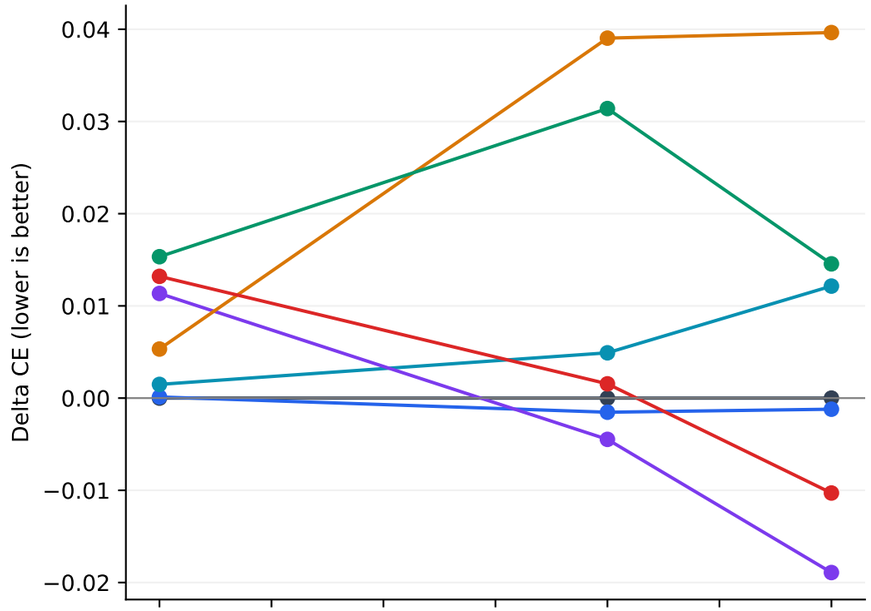
Training loss: trailing 25-step mean



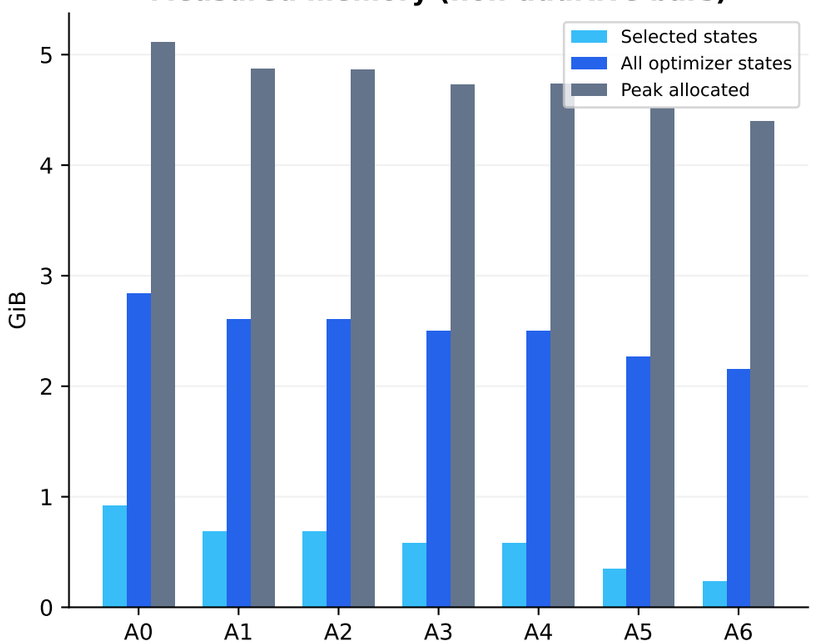
Validation loss: all 3 observations



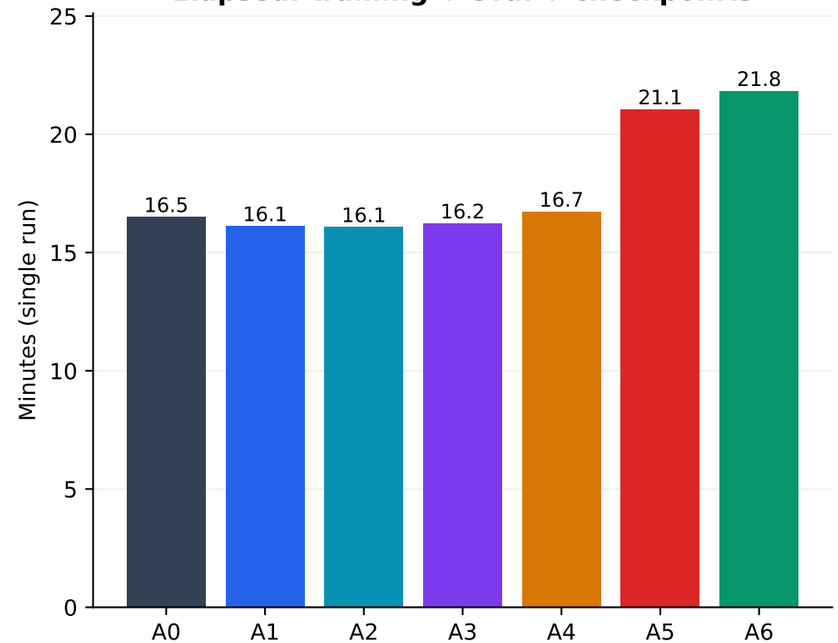
Validation difference vs FP32 baseline



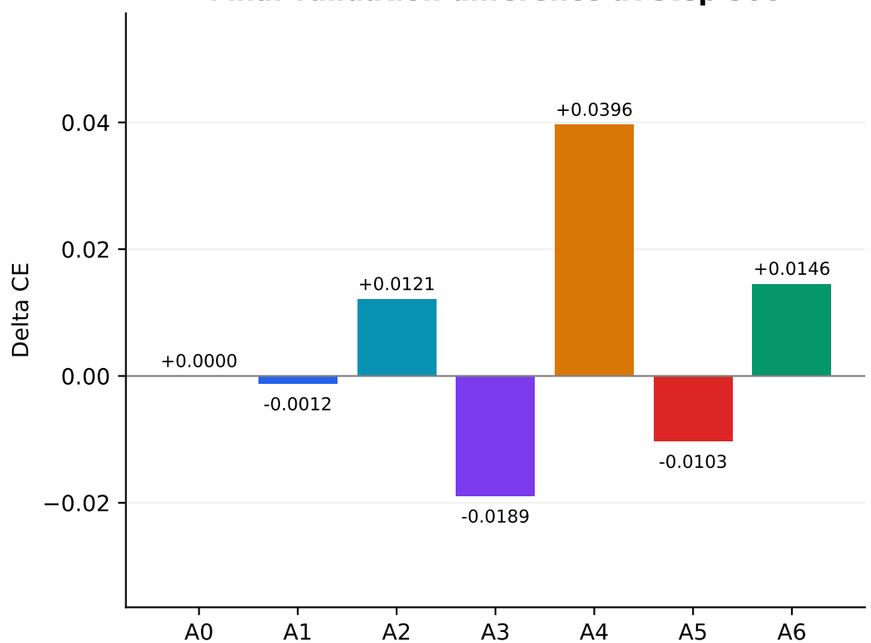
Measured memory (non-additive bars)



Elapsed: training + eval + checkpoints

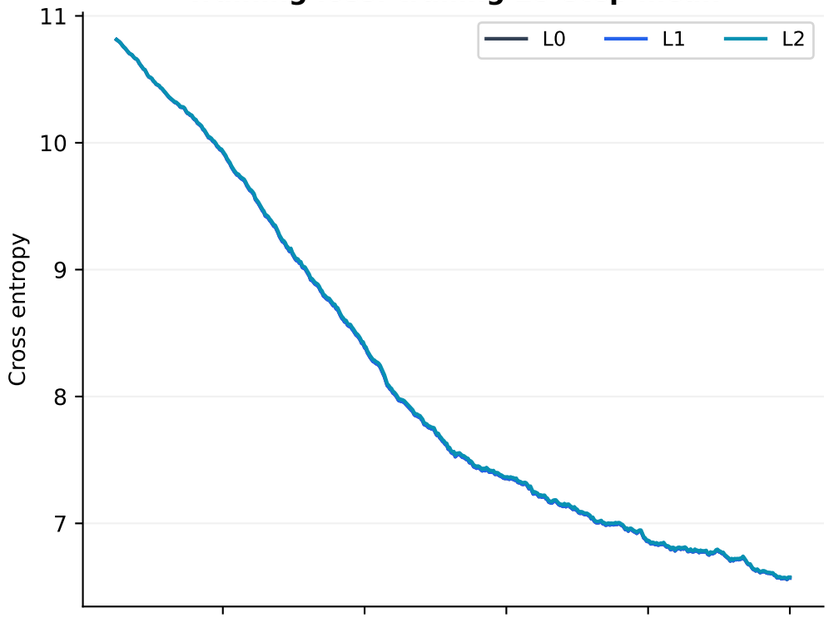


Final validation difference at step 500

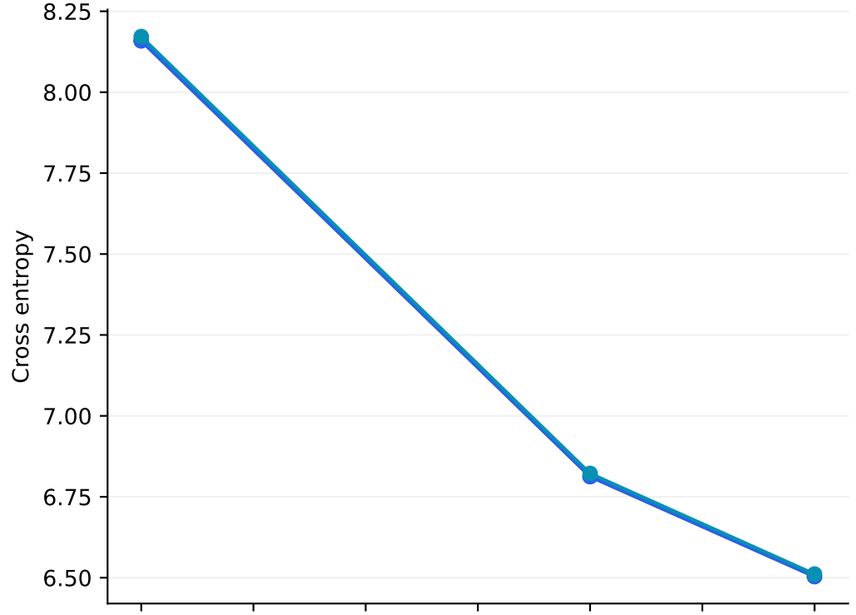


newton-muon | 124M Llama | packed states | 500 steps | seed 0

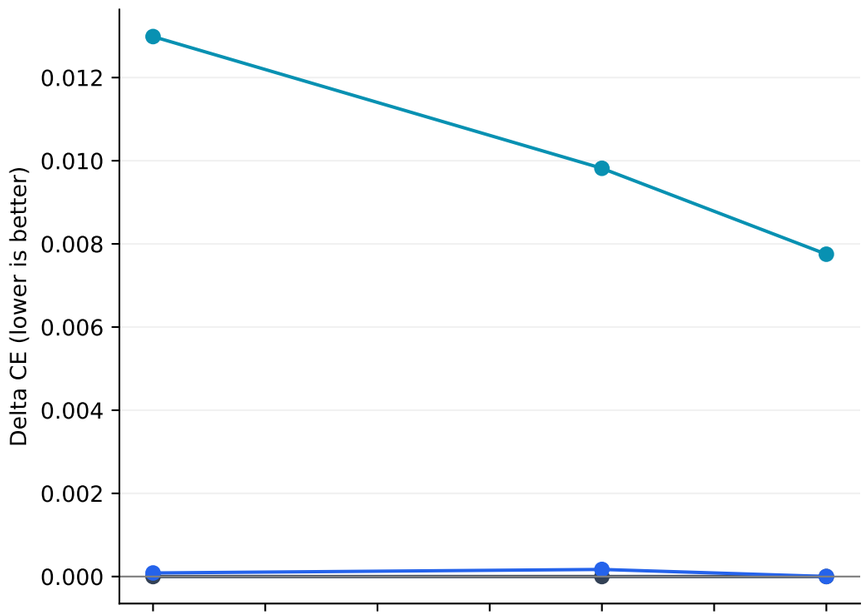
Training loss: trailing 25-step mean



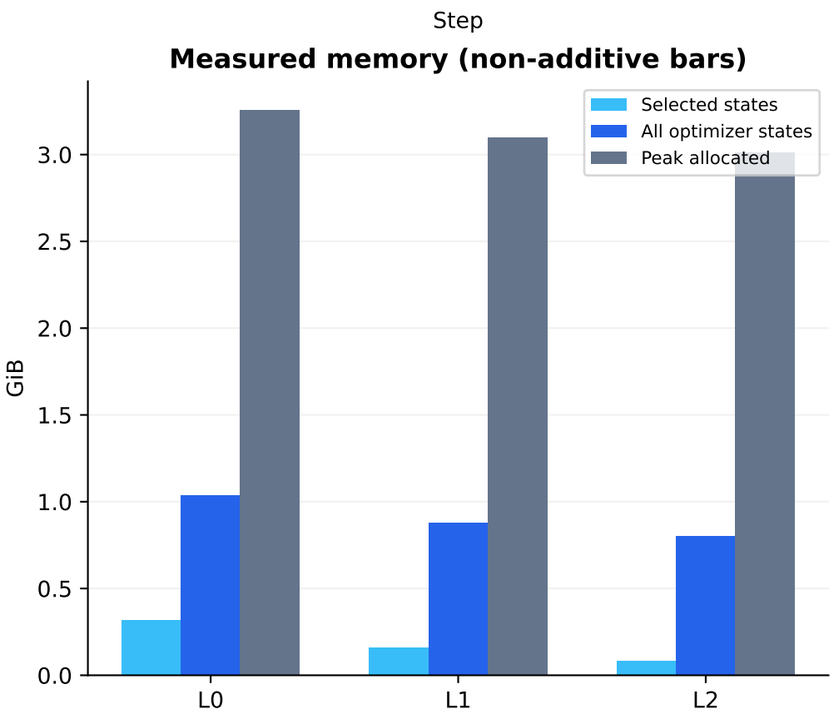
Validation loss: all 3 observations



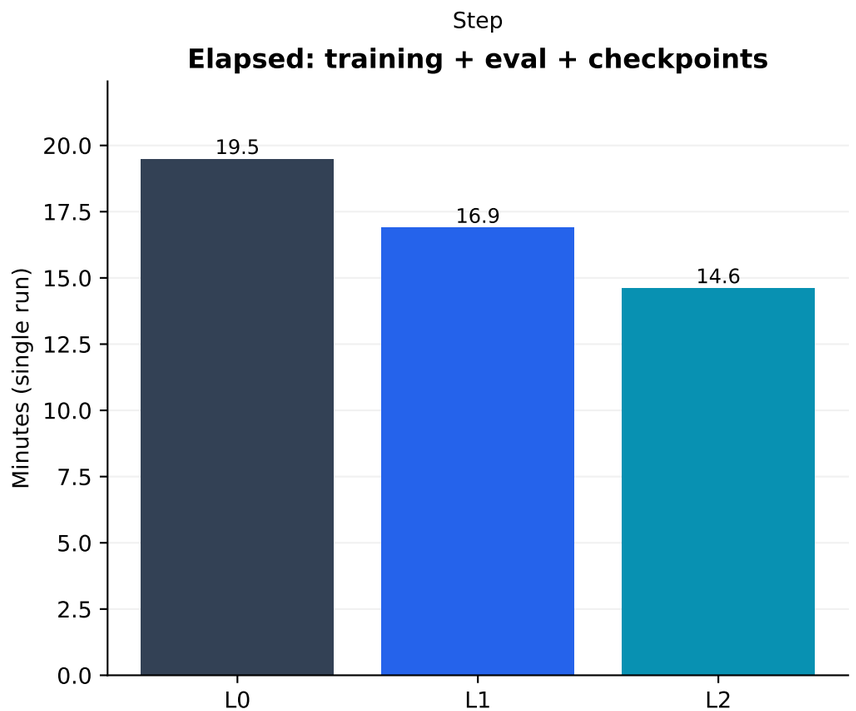
Validation difference vs FP32 baseline



Measured memory (non-additive bars)



Elapsed: training + eval + checkpoints



Final validation difference at step 500

